

High-Dimensional Regularization of the Spatial Sign Covariance Matrix for Robust Shape Estimation

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Abstract—Covariance estimation is a key component of many applications in system identification and data-driven control. Although heavy-tailed distributions may lack a covariance matrix to estimate, the shape matrix provides a well-defined, scale-free generalization for the broad family of elliptical distributions. In this setting, practitioners often employ Tyler’s M-estimator (TME), which is defined implicitly and is computed using a fixed-point iteration. The spatial sign covariance matrix estimator (SSCM) offers a much simpler alternative: it corresponds to one iteration of TME. Yet, the SSCM has been largely regarded as an inferior estimator due to its statistical inconsistency under fixed-dimensional asymptotics. By contrast, using second-order tools from random matrix theory, we establish that, under standard assumptions, SSCM asymptotically dominates TME in Frobenius risk when dimension and sample size grow proportionally. This advantage arises from implicit regularization, which reduces variance at the cost of a bias that vanishes as the dimension grows. Numerical experiments support the theoretical predictions even at moderate dimensions.

I. INTRODUCTION

Covariance matrices characterize the linear dependence structure of random vectors, which, naturally, makes their estimation central to system identification, data-driven control, signal processing, and machine learning [1], [2]. In many practical settings, however, observations are affected by gross outliers and heavy-tailed noise, which greatly complicate estimation and have motivated robust extensions of Kalman filtering [3], methods for training neural networks [4], and data sensitivity measures [5]. With sufficiently heavy tails, the covariance matrix itself may not even exist, leaving sample moments without a corresponding finite population covariance matrix to estimate.

To study dependence in a single framework spanning light- and heavy-tailed noise, we model observations as randomly scaled Gaussian vectors. This representation encompasses many common noise models used in control and state estimation, including the Gaussian itself, as well as Student’s t , elliptically symmetric α -stable, and contaminated Gaussian noise models. Formally, we consider observations of the form

$$\mathbf{X}_i = \boldsymbol{\mu} + \sqrt{W_i} \mathbf{G}_i, \quad i = 1, \dots, n, \quad (1)$$

where each observation $\mathbf{X}_i \in \mathbb{R}^p$ is a column vector whose corresponding $\mathbf{G}_i$ is i.i.d. $N_p(\mathbf{0}, \boldsymbol{\Sigma})$, and the mixing variable

$W_i > 0$ is a scalar random variable. Our goal is to estimate $\boldsymbol{\Sigma}$, the covariance matrix of the latent Gaussian vector $\mathbf{G}_i$, which provides a well-defined estimation target even when the observations $\mathbf{X}_i$ have infinite variance. We make some blanket assumptions following [6]: we suppose that the location vector $\boldsymbol{\mu} \in \mathbb{R}^p$ is known, and without loss of generality set it to $\boldsymbol{\mu} = \mathbf{0}$. Further, we assume $\boldsymbol{\Sigma}$ is positive definite and that $\text{tr}(\boldsymbol{\Sigma}) = p$ for identifiability. Let $\mathbf{X} = [\mathbf{X}_1 \cdots \mathbf{X}_n]^\top$ and $\mathbf{G} = [\mathbf{G}_1 \cdots \mathbf{G}_n]^\top$ denote the $n \times p$ matrices of stacked observations and latent Gaussian vectors, respectively.

Equation (1) includes the elliptical family of distributions, the standard generalization of the Gaussian to heavy-tailed noise commonly used in signal processing and state estimation [7]. Under the previously mentioned trace- p restriction, $\boldsymbol{\Sigma}$ is often called a *shape matrix*, as it defines the scale-free family of ellipsoids that form the level surfaces of the latent Gaussian density. Even when the population covariance matrix does not exist for the observations, if $\mathbf{X}_i$ is itself elliptical and its density exists, the shape matrix likewise defines a representative of its ellipsoidal level surfaces where the p.d.f. remains constant. One such example of the infinite-variance case is the elliptical α -stable distribution with $\alpha < 2$.

While $\mathbf{G}_i$ are themselves assumed to be i.i.d., no independence need be assumed for the mixing variables. Because of this, when the mixing variables are allowed to depend on the Gaussian vectors or on one another, (1) further encompasses substantially non-elliptical distributions. In such cases, $\boldsymbol{\Sigma}$ describes the dependence imposed by the latent Gaussian vectors, but not any additional dependence introduced through the mixing variables. The same separation applies to scale: the trace restriction on $\boldsymbol{\Sigma}$ removes absolute scale, which is instead carried by the mixing variables, but the overall scale of $\mathbf{X}_i$ can be estimated separately or supplied externally, for instance from available conditioning information.

When the law of $\mathbf{X}_i$ is fully specified, maximum likelihood estimators (MLEs) are a natural choice for estimating $\boldsymbol{\Sigma}$, as they are asymptotically consistent and efficient under standard regularity conditions. When the specific distribution of the mixing variable is not known, maximum likelihood can still be used to estimate the shape matrix; because $W_i > 0$, normalizing each observation to unit length removes its random scale, leaving directions distributed according to an angular central Gaussian distribution that depends only on $\boldsymbol{\Sigma}$. One appealing aspect of this approach is that it is robust to model misspecification within the class of models allowed by (1). When it exists, this maximum likelihood estimator for the directions of the normalized vectors is called Tyler’s M-estimator (TME), which is defined as the matrix $\hat{\boldsymbol{\Sigma}}_{\text{TME}}$

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that satisfies the fixed point equation

$$\hat{\Sigma}_{\text{TME}} = \frac{p}{n} \sum_{i=1}^n \frac{\mathbf{X}_i \mathbf{X}_i^\top}{\mathbf{X}_i^\top \hat{\Sigma}_{\text{TME}}^{-1} \mathbf{X}_i}, \quad \text{tr}(\hat{\Sigma}_{\text{TME}}) = p. \quad (2)$$

Among M-estimators for shape on elliptical data, TME minimizes the worst-case asymptotic variance with dimension fixed [8], [9], and is considered by many to be the standard choice for estimating shape. Equation (2) is typically solved iteratively, using the inverse of the current estimate to compute the next step. This procedure becomes expensive in high dimensions, however, where both the inversion and the number of iterations contribute substantially to the cost; even approximating TME cheaply remains an active area of research [10].

A much simpler shape estimator is the spatial sign covariance matrix (SSCM), a robust estimator introduced by [11] and defined as

$$\hat{\Sigma}_{\text{SSCM}} = \frac{p}{n} \sum_{i=1}^n \frac{\mathbf{X}_i \mathbf{X}_i^\top}{\|\mathbf{X}_i\|^2} = \frac{p}{n} \sum_{i=1}^n \frac{\mathbf{X}_i \mathbf{X}_i^\top}{\mathbf{X}_i^\top \mathbf{X}_i}, \quad (3)$$

where $\|\cdot\|$ is the Euclidean norm. The SSCM can be seen as either the sample covariance matrix of Euclidean-normalized observation vectors or as a one-shot version of TME that uses the identity as its initialization. Note that the SSCM is typically trace 1 by construction; in (3) we rescale to trace p for ease of comparison. Both TME and the SSCM are distribution-free for our model, in the sense that the mixing variables, including their dependence, cancel exactly in the fractions of (2) and (3). Likewise, observe that, because $\mathbf{X}_i/\|\mathbf{X}_i\| = \mathbf{G}_i/\|\mathbf{G}_i\|$ when $\boldsymbol{\mu} = \mathbf{0}$, it suffices to consider Gaussian observations, which simplify calculations. While TME is the standard choice for robust shape estimation, the SSCM has effectively been left out of the modern estimation toolbox, in large part because it is statistically inconsistent under classical fixed-dimension asymptotics [12]. Whether that inconsistency is a defect, however, depends on the relationship between dimension and sample size.

Main contribution. We show that the SSCM asymptotically outperforms benchmark approaches in mean-squared error when the dimension p and number of observations n grow to infinity together, and can outperform them at practically relevant finite dimensions. Specifically, the SSCM’s lack of consistency provides beneficial regularization: it trades practically significant reductions in variance for a bias cost that vanishes as the dimension grows.

II. PRELIMINARIES AND ASSUMPTIONS

Using $\|\cdot\|_F$ to denote the Frobenius norm, we quantify the bias–variance tradeoff in shape estimation using Frobenius risk, the expected sum of squared errors across the entries of a shape estimate:

$$\text{Risk}(\hat{\Sigma}, \Sigma) = \mathbb{E} \|\hat{\Sigma} - \Sigma\|_F^2.$$

Where it aids in risk comparison across problems of varying (but finite) dimensions, we normalize this risk by p .

Definition 1 (Asymptotic Dominance). Along a sequence of models, we say that $\hat{\Sigma}_A$ *asymptotically dominates* $\hat{\Sigma}_B$ if

$$\liminf_{n,p \rightarrow \infty} [\text{Risk}(\hat{\Sigma}_B, \Sigma) - \text{Risk}(\hat{\Sigma}_A, \Sigma)] > 0.$$

This comparison retains risk differences that persist asymptotically, even when common leading-order growth makes the risk ratio tend to one. These effects can contribute to material differences in approximate risk at finite dimensions. To study this high-dimensional regime where sample size and dimension grow together, we introduce the following assumptions standard in random matrix theory (RMT). Throughout, the observations satisfy (1) and the conditions stated with it. Note that the numbered assumptions below apply only where explicitly invoked.

Finally, observe that for fixed dimension, Frobenius risk is proportional to the mean-squared error (MSE), and dominance in one implies dominance in the other.

Assumption 1 (High-dimensional regime).

- (a) The dimension $p = p(n)$ grows with the sample size n such that $c_p := p/n \rightarrow c \in (0, \infty)$ as $n, p \rightarrow \infty$.
- (b) The eigenvalues $\lambda_k(\Sigma)$ of Σ are uniformly bounded away from zero such that $\lambda_- \leq \lambda_k(\Sigma) \leq \lambda_+$, $k = 1, \dots, p$, for constants $\lambda_-, \lambda_+ > 0$ independent of p .
- (c) The empirical spectral distribution (ESD) of Σ , which assigns equal weight to each eigenvalue, $H_p(x) := \frac{1}{p} \sum_{k=1}^p \mathbf{1}\{\lambda_k(\Sigma) \leq x\}$ converges weakly to a probability distribution function H as $p \rightarrow \infty$.

Additionally, call $F^{\hat{\Sigma}}$, the ESD of the estimator $\hat{\Sigma}$, defined analogously to H_p . Note that the ESD is defined for deterministic and random matrices alike. All matrices and vectors are understood to belong to sequences indexed by n and p ; to simplify notation, dependence on n and p is suppressed unless needed for clarity. Thus, for example, $\hat{\Sigma}$ and its corresponding ESD $F^{\hat{\Sigma}}$, should be understood as $\hat{\Sigma}^{(p)}$ and $F^{\hat{\Sigma}^{(p)}}$, respectively. For results related to TME, we assume $c_p < 1$ and $c < 1$ so that that estimator exists for every scenario. Finally, we adopt the notation of [13] to denote the moments $\beta_{k,p} := \int x^k dH_p(x) = p^{-1} \text{tr}(\Sigma^k)$, with limit $\beta_{k,p} \xrightarrow{n,p \rightarrow \infty} \beta_k := \int x^k dH(x)$.

We are interested in comparing the performance of three categories of estimators of Σ : unbiased estimators, of which the latent sample covariance matrix $\frac{1}{n} \mathbf{G} \mathbf{G}^\top$ is a well-studied example, the spatial sign covariance matrix, and Tyler’s M-estimator. Analysis of the SSCM and TME, as well as their respective tradeoffs in the conventional setting where p is fixed and $n \rightarrow \infty$ can be found in [12] and [9]. While $\mathbf{G}$ is generally not observable and thus the latent sample covariance matrix is not generally attainable, it provides a useful “oracle” benchmark.

The sample covariance matrix, SSCM, and TME are examples of *rotationally equivariant estimators*. Such estimators, $\hat{\Sigma} = \hat{\Sigma}(\mathbf{X})$, have the property that for any orthogonal matrix $\mathbf{O} \in \mathbb{R}^{p \times p}$, $\hat{\Sigma}(\mathbf{X}\mathbf{O}) = \mathbf{O}^\top \hat{\Sigma}(\mathbf{X}) \mathbf{O}$. Thus, an orthogonal transformation of the data induces the corresponding orthogonal transformation of the estimator, preserving its eigenvalues.

For the Gaussian sample covariance, when p is fixed, its ESD asymptotically converges to that of the population Σ : e.g. as $n \rightarrow \infty$ and $c_p \rightarrow 0$ we have $F^{\hat{\Sigma}} \rightarrow H_p$. For example, when $\Sigma = \mathbf{I}_p$, all sample eigenvalues converge to one almost surely. Under Assumption 1, however, these eigenvalues retain a persistent spread and their ESD converges weakly almost surely to the Marčenko–Pastur (MP) distribution [14], whose continuous part has density

$$f_{\text{MP}}(x) = \frac{\sqrt{(b-x)(x-a)}}{2\pi cx}, \quad a < x < b,$$

where $a = (1 - \sqrt{c})^2$ and $b = (1 + \sqrt{c})^2$. For $c > 1$, the distribution also has an atom of mass $1 - 1/c$ at zero, reflecting the rank deficiency of the sample covariance matrix when $p > n$. Thus, even for an identity population shape, sampling under the regime described by Assumption 1 produces a systematic distortion in the estimated eigenvalues. For general Σ , the limiting spectral distribution of the sample covariance matrix depends on both the aspect ratio c and the population spectrum H . We denote this generalized MP distribution $F^{c,H}$. Mirroring the RMT literature, we also make use of the finite-dimensional proxy F^{c_p, H_p} , a generalized MP distribution corresponding to the finite- p aspect ratio c_p and population ESD H_p ; more details can be found in [15].

For data generated by (1) under Assumption 1, the latent Gaussian sample covariance matrix, SSCM and TME have been found to share a common limiting generalized MP distribution $F^{c,H}$ [6], [13], [15]. This means that when comparing these estimators in the high-dimensional setting, the limiting spectral distribution is insufficient to distinguish between them even when practical differences are large, and second-order tools are required to describe asymptotic dominance.

Our comparisons with unbiased estimators require the following additional restriction on the mixing variables.

Assumption 2 (Gaussian scale mixtures). The mixing vector $(W_1, \dots, W_n)$ is independent of the Gaussian sample $(\mathbf{G}_1, \dots, \mathbf{G}_n)$, and its distribution does not depend on Σ .

When Assumption 2 holds for data generated according to (1), the resulting observations $\mathbf{X}_i$ are frequently referred to as *Gaussian scale mixtures*. Note that unbiasedness here means $\mathbb{E}_{\Sigma}[\hat{\Sigma}] = \Sigma$ for every Σ with $\text{vec}(\Sigma) \in \Phi$, where Φ is the parameter space of Σ , represented in vec coordinates

$$\Phi := \{\text{vec}(\Sigma) : \Sigma \in \mathbb{S}_{++}^p, \text{tr}(\Sigma) = p\},$$

and $\mathbb{S}_{++}^p$ denotes the cone of $p \times p$ real symmetric positive-definite matrices.

Lastly, the following propositions compile some technical details that are useful for proofs of the main results.

Proposition 1 (Uniform Integrability of TME-SSCM Quadratic Trace Gap). *Under the model of (1), suppose Assumption 1 holds with $c_p < 1$ and $c < 1$ and let $\Pi_n := \text{tr}(\hat{\Sigma}_{\text{TME}}^2) - \text{tr}(\hat{\Sigma}_{\text{SSCM}}^2)$. Then $\sup_n \mathbb{E}|\Pi_n|^2 < \infty$ and $\{\Pi_n\}$ is uniformly integrable.*

Proposition 2 (Estimator Bias Control). *Under the model of (1), suppose Assumption 1 holds. Then $\|\mathbb{E}(\hat{\Sigma}_{\text{SSCM}}) - \Sigma\|_F^2 = \mathcal{O}(1/p)$. When we additionally have $c_p, c < 1$, $\|\mathbb{E}(\hat{\Sigma}_{\text{TME}}) - \Sigma\|_F^2 = \mathcal{O}(1/p)$ and $\text{tr}(\mathbb{E}(\hat{\Sigma}_{\text{TME}})\Sigma) = \text{tr}(\Sigma^2) + \mathcal{O}(1/p)$.*

Because Proposition 1 and Proposition 2 are auxiliary to the main results and their proofs require comparatively involved estimator-specific analysis, we provide the details in the online [Supplement](#) to this paper.

III. MAIN RESULTS

Conveniently, for rotationally equivariant estimators, our risk function of interest can be written in terms of linear spectral statistics (LSS).

Definition 2 (Centered Linear Spectral Statistics). Let $\Gamma \in \mathbb{R}^{p \times p}$ be a deterministic positive-semidefinite matrix with ESD K_p . The *centered linear spectral statistics* of an estimator $\hat{\Sigma} \in \mathbb{R}^{p \times p}$ are defined as:

$$L_{\Gamma}^{\hat{\Sigma}}(f) := \int f(x) dL_{\Gamma}^{\hat{\Sigma}}(x) = \int f(x) d[F^{\hat{\Sigma}}(x) - F^{c_p, K_p}(x)]$$

where f is a function on $[0, \infty)$.

Definition 2 follows the conventions of Equation 2.10 in [6]. Next, we write our risk function in terms of LSS.

Lemma 1 (Risk Decomposition). *For any positive-semidefinite rotationally equivariant matrix estimator $\hat{\Sigma}$ with $\mathbb{E}(\hat{\Sigma}) = \Gamma$ and $\text{tr}(\Gamma) = p$, the loss can be expressed as,*

$$\mathbb{E}\|\hat{\Sigma} - \Sigma\|_F^2 = pc_p + p\mathbb{E}[L_{\Gamma}^{\hat{\Sigma}}(x^2)] + \|\Gamma - \Sigma\|_F^2. \quad (4)$$

Proof. Call K_p the ESD of Γ . Then we can write:

$$\begin{aligned} \mathbb{E}\|\hat{\Sigma} - \Sigma\|_F^2 &= \mathbb{E}\text{tr}(\hat{\Sigma}^2) + \text{tr}(\Sigma^2) - 2\text{tr}(\mathbb{E}(\hat{\Sigma})\Sigma) \\ &= p\mathbb{E}\left[\int x^2 dF^{\hat{\Sigma}}(x)\right] + \text{tr}(\Sigma^2) - 2\text{tr}(\Gamma\Sigma) \\ &= p\int x^2 dF^{c_p, K_p}(x) + p\mathbb{E}[L_{\Gamma}^{\hat{\Sigma}}(x^2)] + \text{tr}(\Sigma^2) \\ &\quad - 2\text{tr}(\Gamma\Sigma) \\ &= pc_p + p\mathbb{E}[L_{\Gamma}^{\hat{\Sigma}}(x^2)] + \text{tr}(\Gamma^2) + \text{tr}(\Sigma^2) \\ &\quad - 2\text{tr}(\Gamma\Sigma) \\ &= pc_p + p\mathbb{E}[L_{\Gamma}^{\hat{\Sigma}}(x^2)] + \|\Gamma - \Sigma\|_F^2. \end{aligned}$$

□

The decomposition in Lemma 1 is primarily a bookkeeping device: it further separates the variance term in the bias-variance decomposition of MSE into an irreducible contribution from the Marčenko–Pastur limit, plus a deviation, $\mathbb{E}(L_{\Gamma}^{\hat{\Sigma}}(x^2))$. Note that this is an exact result; even for finite n, p , $\mathbb{E}(L_{\Gamma}^{\hat{\Sigma}}(x^2))$ provides a way to compare the pre-asymptotic risk of estimators that share a limiting spectral distribution.

In special cases, this expectation can be derived analytically. In general, however, the exact distribution of $L_{\Gamma}^{\hat{\Sigma}}(x^2)$ —

and thus its expectation — are unknown for an arbitrary estimator. For certain estimators, central limit theorems for linear spectral statistics provide a way to quantify how such estimators compare in the high-dimensional limit.

In particular, [15] derives a central limit theorem for the linear spectral statistics of sample covariance matrices on certain distributions. These were recently extended to a central limit theorem for LSS of the SSCM by [6], and to those of Tyler's M-estimator in [13]. These papers employ a deterministic centering matrix, specialized to (1) as

$$\mathbf{T} := \Sigma - \frac{2}{p} \left(\Sigma^2 - \beta_{2,p} \Sigma \right),$$

to account for the nonlinearities induced by the denominators of the SSCM and TME. In particular, [6] show that $\mathbf{T}$ approximates $\Gamma_{\text{SSCM}} := \mathbb{E}(\hat{\Sigma}_{\text{SSCM}})$ to order $o(p^{-1})$ in operator norm, which we denote $\|\cdot\|_{\text{op}}$ throughout. Finally, define the TME population mean similarly as $\Gamma_{\text{TME}} := \mathbb{E}(\hat{\Sigma}_{\text{TME}})$.

For the spatial sign covariance matrix and Tyler's M-estimator, we can quantify the LSS means of Lemma 1 using the following lemma.

Lemma 2 (Naturally-Centered LSS Means for SSCM and Tyler's M-Estimator). *Suppose we have data generated by the model defined in (1). Then, the following hold true.*

- a) *For the SSCM, for any n, p , $p\mathbb{E}(L_{\Gamma_{\text{SSCM}}}^{\hat{\Sigma}_{\text{SSCM}}}(x^2)) = -\frac{1}{n}\text{tr}(\Gamma_{\text{SSCM}}^2)$. When Assumption 1 holds, $p\mathbb{E}(L_{\Gamma_{\text{SSCM}}}^{\hat{\Sigma}_{\text{SSCM}}}(x^2)) \rightarrow -c\beta_2$ as $n, p \rightarrow \infty$;*
- b) *Suppose Assumption 1 holds with $c_p, c < 1$ so that Tyler's M-estimator exists. Then $p\mathbb{E}(L_{\Gamma_{\text{TME}}}^{\hat{\Sigma}_{\text{TME}}}(x^2)) \rightarrow c\beta_2 + \frac{2c}{(1-c)}$ as $n, p \rightarrow \infty$.*

Proof. For the SSCM, let $\mathbf{Y}_i = \mathbf{X}_i/\|\mathbf{X}_i\|$ and $\Gamma_{\text{SSCM}} = p\mathbb{E}(\mathbf{Y}_i \mathbf{Y}_i^\top)$. Using $\|\mathbf{Y}_i\| = 1$ and independent observations,

$$\mathbb{E} \text{tr}(\hat{\Sigma}_{\text{SSCM}}^2) = pc_p + \left(1 - \frac{1}{n}\right) \text{tr}(\Gamma_{\text{SSCM}}^2).$$

Matching terms with Lemma 1 therefore gives

$$p\mathbb{E} \left[L_{\Gamma_{\text{SSCM}}}^{\hat{\Sigma}_{\text{SSCM}}}(x^2) \right] = -\frac{1}{n} \text{tr}(\Gamma_{\text{SSCM}}^2) \xrightarrow{n, p \rightarrow \infty} -c\beta_2,$$

where the limit follows from $\|\Gamma_{\text{SSCM}} - \mathbf{T}\|_{\text{op}} = o(p^{-1})$ and $\mathbf{T} = \Sigma + \mathcal{O}(p^{-1})$ under the bounded-spectrum assumption [6], so that

$$\begin{aligned} \text{tr}(\Gamma_{\text{SSCM}}^2) &= \text{tr}(\mathbf{T}^2) + o(1) \\ &= \text{tr}(\Sigma^2) + \frac{4}{p} [\beta_{2,p} \text{tr}(\Sigma^2) - \text{tr}(\Sigma^3)] + o(1) \\ &= \text{tr}(\Sigma^2) + 4(\beta_{2,p}^2 - \beta_{3,p}) + o(1). \end{aligned}$$

Observe that under this construction, we have a population-centered LSS mean for the SSCM given by

$$\begin{aligned} \mathbb{E} \text{tr}(\hat{\Sigma}_{\text{SSCM}}^2) - \text{tr}(\Sigma^2) - pc_p \\ &= -\frac{1}{n} \text{tr}(\Gamma_{\text{SSCM}}^2) + \text{tr}(\Gamma_{\text{SSCM}}^2) - \text{tr}(\Sigma^2) \\ &\xrightarrow{n, p \rightarrow \infty} -c\beta_2 + 4(\beta_2^2 - \beta_3). \end{aligned}$$

For TME, Proposition 1 implies that the limiting value of Π_n given by ([13], S5.3) holds in expectation:

$$\mathbb{E} \text{tr}(\hat{\Sigma}_{\text{TME}}^2) - \mathbb{E} \text{tr}(\hat{\Sigma}_{\text{SSCM}}^2) \xrightarrow{n, p \rightarrow \infty} \frac{2c}{1-c} + 2c\beta_2 + 4(\beta_3 - \beta_2^2).$$

Adding this to the population-centered LSS mean for the SSCM, we get

$$\mathbb{E} \text{tr}(\hat{\Sigma}_{\text{TME}}^2) - \text{tr}(\Sigma^2) - pc_p \xrightarrow{n, p \rightarrow \infty} c\beta_2 + \frac{2c}{1-c}.$$

Then by Lemma 1, $p\mathbb{E}[L_{\Gamma_{\text{TME}}}^{\hat{\Sigma}_{\text{TME}}}(x^2)] = \mathbb{E} \text{tr}(\hat{\Sigma}_{\text{TME}}^2) - pc_p - \text{tr}(\Gamma_{\text{TME}}^2)$. Since Proposition 2 gives $\|\mathbb{E}(\hat{\Sigma}_{\text{TME}}) - \Sigma\|_F^2 = \mathcal{O}(1/p)$ and $\text{tr}(\mathbb{E}(\hat{\Sigma}_{\text{TME}})\Sigma) = \text{tr}(\Sigma^2) + \mathcal{O}(1/p)$, the result follows. $\square$

The following result shows that, in a finite n, p setting, the SSCM can achieve lower risk than any unbiased shape estimator for a Gaussian scale mixture.

Theorem 1 (Exact Frobenius Risk Dominance of SSCM Over Unbiased Estimators For GSM). *For a sample generated by (1) under Assumption 2, let $\hat{\Sigma}$ be a symmetric matrix-valued estimator of Σ that is unbiased on Φ . Further, assume that the Fisher information matrix for the shape matrix exists. If $\|\Gamma_{\text{SSCM}} - \Sigma\|_F^2 \leq \frac{1}{n} \text{tr}(\Gamma_{\text{SSCM}}^2) + \frac{1}{n} \text{tr}(\Sigma^2) - \frac{2}{n} \text{tr}(\Sigma^4) \text{tr}(\Sigma^2)^{-1}$, then $\mathbb{E}\|\hat{\Sigma}_{\text{SSCM}} - \Sigma\|_F^2 \leq \mathbb{E}\|\hat{\Sigma} - \Sigma\|_F^2$. The SSCM's risk advantage over any such estimator is at least the margin by which this condition is satisfied.*

The proof of Theorem 1, provided in the Appendix, uses a Cramér–Rao bound under the trace constraint. This bound involves the Fisher information matrix, defined as the expected outer product of the gradient of the sample log-likelihood with respect to the shape parameters. Intuitively, Theorem 1 shows that the SSCM's variance advantage can compensate for the squared bias it incurs when Σ is not equal to $\mathbf{I}_p$ (where the SSCM is unbiased). Although the theorem's condition involves unknown population quantities, under Assumption 1 its left-hand side is $\mathcal{O}(p^{-1})$, while its right-hand side is $\mathcal{O}(1)$. Practically speaking, this condition is satisfied fairly rapidly in moderate dimensional problems. An immediate consequence is that the SSCM asymptotically dominates every unbiased estimator irrespective of the true shape matrix.

Corollary 1 (Limiting Frobenius Risk Dominance of SSCM Over Unbiased Estimators For GSM). *Consider a sample generated by (1), and suppose Assumptions 1 and 2 hold. Let $\hat{\Sigma}$ be a symmetric matrix-valued estimator of Σ that is unbiased on Φ . Further, suppose that the Fisher information matrix for the shape matrix exists. Then $\hat{\Sigma}_{\text{SSCM}}$ asymptotically dominates every unbiased estimator $\hat{\Sigma}$.*

Proof. This follows from Theorem 1; the conclusion of the theorem gives the following lower bound on the risk advantage:

IV. SIMULATION RESULTS

$$\begin{aligned} & \mathbb{E}\|\hat{\Sigma} - \Sigma\|_F^2 - \mathbb{E}\|\hat{\Sigma}_{\text{SSCM}} - \Sigma\|_F^2 \\ & \geq \frac{1}{n}\text{tr}(\Sigma^2) + \frac{1}{n}\text{tr}(\Gamma_{\text{SSCM}}^2) - \frac{2}{n}\frac{\text{tr}(\Sigma^4)}{\text{tr}(\Sigma^2)} \\ & \quad - \|\Gamma_{\text{SSCM}} - \Sigma\|_F^2. \end{aligned}$$

By definition, $p^{-1}\text{tr}(\Sigma^2) \rightarrow \beta_2$ and $p^{-1}\text{tr}(\Gamma_{\text{SSCM}}^2) = p^{-1}\text{tr}(\Sigma^2) + o(1) \rightarrow \beta_2$, while the remaining terms are $o(1)$ by the bounded spectrum assumption and Proposition 2. Therefore, the SSCM risk advantage is bounded by

$$\liminf_{n,p \rightarrow \infty} \left[\mathbb{E}\|\hat{\Sigma} - \Sigma\|_F^2 - \mathbb{E}\|\hat{\Sigma}_{\text{SSCM}} - \Sigma\|_F^2 \right] \geq 2c\beta_2 > 0.$$

Hence, in the high-dimensional limit, the SSCM has strictly smaller Frobenius risk than every unbiased estimator $\hat{\Sigma}$. Note that the limit here is a $\liminf$, since no convergence assumption is made on the risk of an arbitrary unbiased estimator. $\square$

The classical appeal of MLEs stems in large part from their unbiasedness and efficiency under standard fixed-dimensional asymptotics. In the high-dimensional regime considered here, the SSCM's squared bias likewise vanishes, while its regularization delivers a persistent variance advantage over every unbiased estimator. Compared with MLEs based on a fully specified likelihood function for $\mathbf{X}_i$, the SSCM also avoids the associated risk of misspecification of the mixing law.

Next, we show that this SSCM outperformance carries over to Tyler's M-estimator with even looser model assumptions.

Theorem 2 (High-Dimensional Frobenius Risk Dominance of SSCM Over Tyler's M-Estimator). *Consider a sample generated by (1), and suppose Assumption 1 holds, with $c < 1$. Then $\hat{\Sigma}_{\text{SSCM}}$ asymptotically dominates $\hat{\Sigma}_{\text{TME}}$.*

Proof.

$$\begin{aligned} & \mathbb{E}\|\hat{\Sigma}_{\text{TME}} - \Sigma\|_F^2 - \mathbb{E}\|\hat{\Sigma}_{\text{SSCM}} - \Sigma\|_F^2 \\ & = pc_p + p\mathbb{E}[L_{\Gamma_{\text{TME}}}^{\hat{\Sigma}_{\text{TME}}}(x^2)] + \|\Gamma_{\text{TME}} - \Sigma\|_F^2 \\ & \quad - \left(pc_p + p\mathbb{E}[L_{\Gamma_{\text{SSCM}}}^{\hat{\Sigma}_{\text{SSCM}}}(x^2)] + \|\Gamma_{\text{SSCM}} - \Sigma\|_F^2 \right) \\ & = p\mathbb{E}[L_{\Gamma_{\text{TME}}}^{\hat{\Sigma}_{\text{TME}}}(x^2)] - p\mathbb{E}[L_{\Gamma_{\text{SSCM}}}^{\hat{\Sigma}_{\text{SSCM}}}(x^2)] + \mathcal{O}(1/p) \\ & \xrightarrow{n,p \rightarrow \infty} c\beta_2 + \frac{2c}{(1-c)} + c\beta_2 \\ & = 2c\beta_2 + \frac{2c}{(1-c)} > 0. \end{aligned}$$

In the above, the risk as a function of LSS means is due to Lemma 1. The Frobenius bias for both SSCM and TME is vanishing as a result of Proposition 2, and the limiting values of the centered LSS means are provided in Lemma 2. $\square$

Theorem 2 applies to shape matrix estimation in the more general model of (1), not merely the special case of Gaussian scale mixtures required by Assumption 2.

In this section, we provide numerical simulation results that demonstrate the accuracy of the estimation errors predicted by Lemma 1, using the limiting values of the naturally-centered means of LSS provided in Lemma 2, even when the dimension is only moderately large. In doing so, we intentionally leave out the finite- p bias-correction from the SSCM, emphasizing its small penalty even in problems far from the asymptotic regime.

Because the SSCM and TME are both distribution-free in the sense that each estimator is invariant to the choice of the mixing variable W_i in (1), we set $W_i = 1$ for $i = 1, \dots, n$ and simulate from a multivariate Gaussian with mean zero and a trace- p covariance matrix parameterized as follows: the dimension p , a spike fraction ϵ , and a spike magnitude $\rho > 1$. For each scenario, we form a diagonal covariance matrix with $\lceil \epsilon p \rceil$ eigenvalues equal to ρ and the remaining eigenvalues equal to 1, followed by normalization to enforce $\text{tr}(\Sigma) = p$. Within each scenario, every estimator is computed from the same 150 simulated samples, so comparisons between estimators are paired: risk differences are averaged replicate by replicate, and their Monte Carlo standard errors account for the sampling variation the estimators share. Standard errors for simulated risk are given in the [Supplement](#).

In addition to the SSCM and TME, we also benchmark the sample covariance matrix on this Gaussian data, which can be thought of as simulating the unobservable latent Gaussian vector in (1). This is not a perfectly matched benchmark for shape estimation, as the sample covariance must also estimate an overall scale while TME and the SSCM are trace- p by construction. Nevertheless, we use the unnormalized sample covariance as a benchmark because it is the canonical covariance estimator in the Gaussian setting, and provides a familiar reference point for covariance estimation. Additionally, because it exploits the Gaussian structure of the latent variables in (1), it can be thought of as an “oracle” benchmark. Although the additional scale estimation problem may appear to give the sample covariance a disadvantage, this distinction becomes asymptotically negligible in the high-dimensional regime considered here. We scale Frobenius errors by $1/p$ to make cleaner comparisons between scenarios of different (finite) dimensions.

Fig. 1 shows the limiting theoretical and average simulated Frobenius risk over 150 draws as $c_p = p/n$ varies, while Fig. 2 depicts the same general simulation setup but with $c_p = 0.5$ fixed and varying the dimension p . These plots show that even at a relatively low n, p , simulated errors coincide with their high-dimensional limiting values remarkably well. The separation between estimators is far larger than simulation variability: at every plotted point in Fig. 1 and Fig. 2, the paired difference in risk between the SSCM and each benchmark exceeds 12 Monte Carlo standard errors. We provide a tabular summary of additional simulation scenarios in the online [Supplement](#) to this paper.

As one would expect from the fact that the sample covariance matrix, SSCM, and TME share a limiting spectral

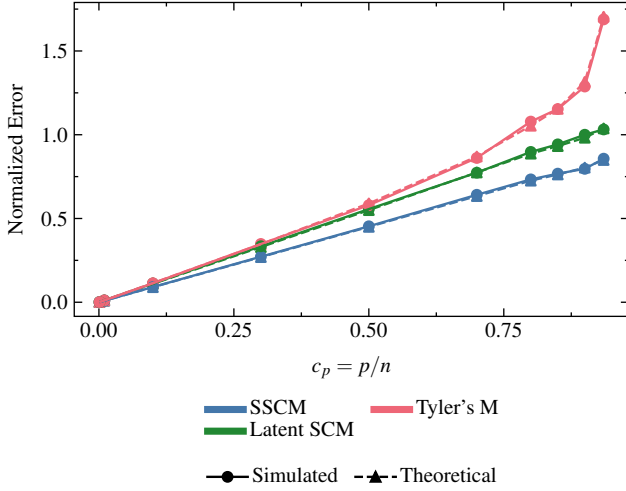


Fig. 1. Simulated vs. high-dimensional theoretical normalized error, $p^{-1}\mathbb{E}|\hat{\Sigma} - \Sigma|_F^2$, for $p = 50$, $\rho = 10,000$, $\epsilon = 0.2$, and various values of c_p . Simulated values average 150 replicates per configuration. Latent SCM denotes the latent Gaussian oracle sample covariance matrix used as a benchmark.

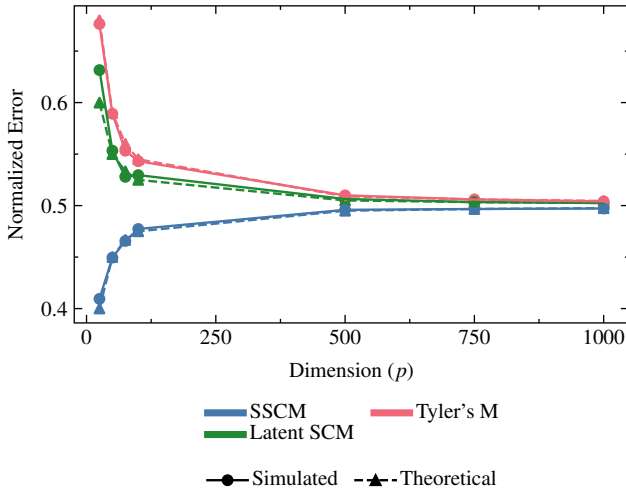


Fig. 2. Simulated vs. theoretical normalized error, $p^{-1}\mathbb{E}|\hat{\Sigma} - \Sigma|_F^2$, for $c_p = 0.5$, $\rho = 10,000$, $\epsilon = 0.2$, and various values of p . The plots above show the average risk over 150 simulations for each configuration.

distribution, Fig. 2 demonstrates that the normalized risk of shape matrix estimates converges to the same value as $n, p \rightarrow \infty$. Importantly, however, this figure clearly shows that as a consequence of its implicit regularization, the SSCM converges from below. Likewise, the relative benefits of SSCM's regularization are the greatest in moderate dimensions, where the LSS error term is not yet dwarfed by the total risk of the estimator (i.e., the other terms in (4)). For example, at $(p, n, \epsilon, \rho) = (25, 250, 0.2, 100)$, the SSCM provides a 29.1% (SE 1.6%) and 30.2% (SE 1.2%) improvement over the latent sample covariance and TME respectively, with an order-of-magnitude runtime reduction over TME.

Conversely, when the dimension is large enough that the

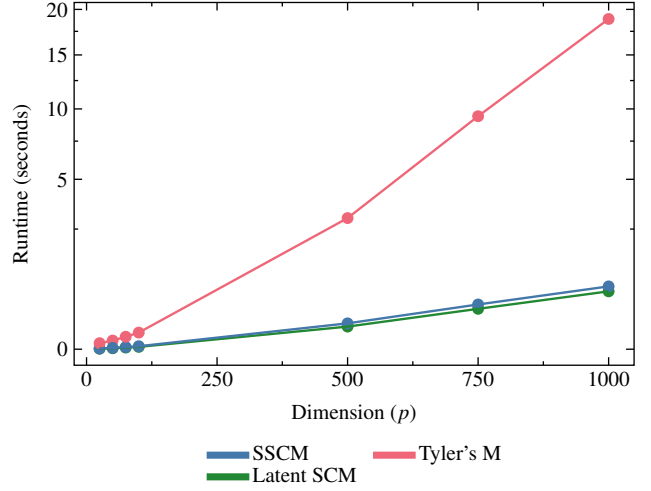


Fig. 3. Estimator runtime comparisons for $\rho = 10,000$, $\epsilon = 0.2$ at various dimensions, averaged over values of c_p .

differences are not practically significant, the computational benefits of the SSCM versus more complex algorithms such as TME are material. Example computational costs for each estimator in problems of increasing dimensionality are shown in Fig. 3.

V. CONCLUSION

Using the criterion in Definition 1, we show that the SSCM asymptotically dominates every unbiased shape estimator for Gaussian scale mixtures, as well as Tyler's M-estimator under the broader model in (1). These improvements in Frobenius risk arise from implicit regularization of the SSCM in the high-dimensional regime: its variance reduction persists while the squared bias vanishes as dimension grows.

Simulation studies show that the theoretical limiting risk is remarkably accurate even when the ambient dimension of the data is only moderate. Moreover, the comparatively cheap computational cost of the SSCM in high dimensions makes it an appealing choice even in situations where the theoretical performance difference is insignificant in practical terms.

While Corollary 1 shows that the Spatial Sign Covariance Matrix dominates all unbiased estimators in the high-dimensional limit, we must stress that this does not imply dominance over all *asymptotically* unbiased estimators such as MLEs, which may be biased in finite samples and can attain better efficiency than the Cramér-Rao lower bound allows without violating any theoretical laws. Additionally, unbiasedness under fixed- p asymptotics does not guarantee unbiasedness in the $n, p \rightarrow \infty$ regime we study. There is currently no theoretical framework to establish the general dominance of the SSCM over MLEs in the high-dimensional limit. That said, improving upon MLE performance for heavy-tailed elliptical vectors was a primary motivation for this research, and empirical results against the benchmarks presented in the previous section show that the SSCM can and does outperform MLEs in the settings studied here. These results suggest that the SSCM merits serious consideration

by practitioners as an alternative to likelihood-based shape estimators.

APPENDIX: PROOF OF THEOREM 1

Proof. The proof proceeds in three steps. First, we show that the latent Gaussian model yields a Cramér-Rao bound no larger than that of any corresponding Gaussian scale mixture. We then derive this bound for the shape matrix and finally compare the resulting risk to that of the SSCM. Because the CRB bounds the risk of any unbiased estimator, any conditions that result in SSCM risk lower than this benchmark risk also imply that SSCM dominates all unbiased estimators of Σ .

Let $\hat{\Sigma}$ be a symmetric matrix-valued estimator of Σ that is unbiased on Φ . Recall that under (1),

$$\mathbf{G}_i \sim N_p(\mathbf{0}, \Sigma) \text{ and } \text{tr}(\Sigma) = p,$$

so the shape matrix of interest is also exactly the covariance matrix of the latent Gaussian vector, which is independent of W_i by Assumption 2.

To obtain the Gaussian Cramér-Rao bound for the trace- p shape matrix, we start with a standard unrestricted covariance parameterization in the Gaussian model and apply the equality-constrained Cramér-Rao bound of [16]. In addition to the trace- p parameter space Φ defined in the problem formulation, we introduce the ambient unrestricted parameter space represented in (nonredundant) half-vectorization coordinates:

$$\Theta := \{ \text{vech}(\Sigma) : \Sigma \in \mathbb{S}_{++}^p \}.$$

We write $\theta \in \Theta$ for the unrestricted parameter vector and $\phi \in \Phi$ for its trace-constrained counterpart. It is convenient to represent the unrestricted parameter in the nonredundant vech coordinates in $\mathbb{R}^{p(p+1)/2}$, while expressing the constrained shape parameter in vec coordinates in $\mathbb{R}^{p^2}$; to bridge these representations, let $\mathbf{D}_p$ denote the duplication matrix, so that

$$\text{vec}(\Sigma) = \mathbf{D}_p \theta.$$

The trace constraint can be written as

$$a(\theta) = \text{tr}(\Sigma) - p = \text{vec}(\mathbf{I}_p)^\top \mathbf{D}_p \theta - p = 0,$$

and its gradient is therefore

$$\nabla_{\theta} a(\theta) = \mathbf{D}_p^\top \text{vec}(\mathbf{I}_p).$$

By [16], there exists $\mathbf{U}$ with orthonormal columns spanning the nullspace of the constraint gradient, so that

$$\nabla_{\theta} a(\theta)^\top \mathbf{U} = \mathbf{0} \text{ and } \mathbf{U}^\top \mathbf{U} = \mathbf{I}.$$

Let $\mathcal{I}_{\mathbf{X}}$ and $\mathcal{I}_{\mathbf{G}}$ denote the Fisher Information Matrices (FIMs) with respect to θ based on the observed Gaussian scale mixture and latent Gaussian sample, respectively. Since W_i itself carries no information about Σ , while $\mathbf{G}_i = \mathbf{X}_i / \sqrt{W_i}$ is recovered from the complete data $(\mathbf{X}_i, W_i)$, the missing-information principle [17] gives

$$\mathcal{I}_{\mathbf{G}} = \mathcal{I}_{\mathbf{X}} + \mathbb{E}_{\mathbf{X}} [\text{Var}(\nabla_{\theta} \log f_{\mathbf{X}|W}(\mathbf{X} | W, \theta) | \mathbf{X})] \succeq \mathcal{I}_{\mathbf{X}}.$$

By Theorem 1 of [16], the equality-constrained CRB matrices for the vectorized shape parameter are

$$\text{CRB}_{\phi, \mathbf{G}} = \mathbf{D}_p \mathbf{U} (\mathbf{U}^\top \mathcal{I}_{\mathbf{G}} \mathbf{U})^{-1} \mathbf{U}^\top \mathbf{D}_p^\top, \text{ and}$$

$$\text{CRB}_{\phi, \mathbf{X}} = \mathbf{D}_p \mathbf{U} (\mathbf{U}^\top \mathcal{I}_{\mathbf{X}} \mathbf{U})^{-1} \mathbf{U}^\top \mathbf{D}_p^\top.$$

This implies that

$$\mathbf{U}^\top \mathcal{I}_{\mathbf{G}} \mathbf{U} \succeq \mathbf{U}^\top \mathcal{I}_{\mathbf{X}} \mathbf{U},$$

where inversion reverses the positive-semidefinite ordering, and congruence by $\mathbf{D}_p \mathbf{U}$ preserves it. Hence

$$\text{CRB}_{\phi, \mathbf{G}} \preceq \text{CRB}_{\phi, \mathbf{X}}.$$

Finally, we obtain the relevant Gaussian bound explicitly. For n independent Gaussian observations with covariance Σ , the FIM under the nonredundant parameterization $\theta = \text{vech}(\Sigma)$ is

$$\mathcal{I}_{\mathbf{G}} = \frac{n}{2} \mathbf{D}_p^\top (\Sigma^{-1} \otimes \Sigma^{-1}) \mathbf{D}_p,$$

which is nonsingular [18]. Thus, defining the unrestricted Gaussian covariance bound by

$$\text{CRB}_{\theta, \mathbf{G}} = \mathcal{I}_{\mathbf{G}}^{-1},$$

Corollary 1 of [16] and the transformation from θ to ϕ give

$$\begin{aligned} \text{CRB}_{\phi, \mathbf{G}} = \mathbf{D}_p & \left(\text{CRB}_{\theta, \mathbf{G}} - \text{CRB}_{\theta, \mathbf{G}} \nabla_{\theta} a(\theta) \right. \\ & \times \left[\nabla_{\theta} a(\theta)^\top \text{CRB}_{\theta, \mathbf{G}} \nabla_{\theta} a(\theta) \right]^{-1} \\ & \left. \times \nabla_{\theta} a(\theta)^\top \text{CRB}_{\theta, \mathbf{G}} \right) \mathbf{D}_p^\top. \end{aligned} \quad (5)$$

The first term is simply the unrestricted CRB matrix represented in vec coordinates:

$$\mathbf{D}_p \text{CRB}_{\theta, \mathbf{G}} \mathbf{D}_p^\top = \frac{1}{n} (\mathbf{I}_{p^2} + \mathcal{K}) (\Sigma \otimes \Sigma),$$

where $\mathcal{K}$ is the commutation matrix [18]. Using $\nabla_{\theta} a(\theta) = \mathbf{D}_p^\top \text{vec}(\mathbf{I}_p)$ then gives

$$\mathbf{D}_p \text{CRB}_{\theta, \mathbf{G}} \nabla_{\theta} a(\theta) = \frac{2}{n} \text{vec}(\Sigma^2), \text{ and}$$

$$\nabla_{\theta} a(\theta)^\top \text{CRB}_{\theta, \mathbf{G}} \nabla_{\theta} a(\theta) = \frac{2}{n} \text{tr}(\Sigma^2).$$

Plugging these into (5), we arrive at

$$\text{CRB}_{\phi, \mathbf{G}} = \frac{1}{n} (\mathbf{I}_{p^2} + \mathcal{K}) (\Sigma \otimes \Sigma) - \frac{2}{n \text{tr}(\Sigma^2)} \text{vec}(\Sigma^2) \text{vec}(\Sigma^2)^\top.$$

For any unbiased estimator based on Gaussian scale mixture observations, the Cramér-Rao bound and the ordering above imply

$$\text{Cov}(\text{vec}(\hat{\Sigma})) \succeq \text{CRB}_{\phi, \mathbf{X}} \succeq \text{CRB}_{\phi, \mathbf{G}}.$$

Additionally, because $\hat{\Sigma}$ is unbiased, the Frobenius risk equals the total variance of its vectorized entries. Taking traces therefore gives the risk bound for unbiased estimators as

$$\begin{aligned}\mathbb{E}\|\hat{\Sigma} - \Sigma\|_F^2 &= \text{tr}\left[\text{Cov}\left(\text{vec}(\hat{\Sigma})\right)\right] \\ &\geq \text{tr}(\text{CRB}_{\phi, \mathbf{G}}) \\ &= pc_p + \frac{1}{n} \text{tr}(\Sigma^2) - \frac{2}{n} \frac{\text{tr}(\Sigma^4)}{\text{tr}(\Sigma^2)}.\end{aligned}$$

Next, let $\mathbf{\Gamma}_{\text{SSCM}} = \mathbb{E}(\hat{\Sigma}_{\text{SSCM}})$. By Lemma 1 and Lemma 2, the SSCM has exact Frobenius risk given by

$$\mathbb{E}\|\hat{\Sigma}_{\text{SSCM}} - \Sigma\|_F^2 = pc_p - \frac{1}{n} \text{tr}(\mathbf{\Gamma}_{\text{SSCM}}^2) + \|\mathbf{\Gamma}_{\text{SSCM}} - \Sigma\|_F^2.$$

Subtracting this expression from the preceding lower bound gives the quantitative risk advantage stated in the theorem. Therefore, the SSCM has no greater risk than the Gaussian Cramér–Rao lower bound whenever

$$\begin{aligned}pc_p - \frac{1}{n} \text{tr}(\mathbf{\Gamma}_{\text{SSCM}}^2) + \|\mathbf{\Gamma}_{\text{SSCM}} - \Sigma\|_F^2 &\leq \\ pc_p + \frac{1}{n} \text{tr}(\Sigma^2) - \frac{2}{n} \frac{\text{tr}(\Sigma^4)}{\text{tr}(\Sigma^2)}.\end{aligned}$$

Rearranging yields

$$\begin{aligned}\|\mathbf{\Gamma}_{\text{SSCM}} - \Sigma\|_F^2 &\leq \\ \frac{1}{n} \text{tr}(\mathbf{\Gamma}_{\text{SSCM}}^2) + \frac{1}{n} \text{tr}(\Sigma^2) - \frac{2}{n} \frac{\text{tr}(\Sigma^4)}{\text{tr}(\Sigma^2)},\end{aligned}$$

which is exactly the condition stated in Theorem 1. $\square$

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Supplement

This online supplement collects the proofs of Proposition 1 and Proposition 2, as well as tables of simulation results.

S1. PROOF OF PROPOSITION 1

The machinery needed to prove the uniform integrability of $\{\Pi_n\}$ is provided in the Supplement to [13]; where possible, we introduce new objects here borrowing their notation. Throughout, the standing conditions of (1) and Assumption 1 hold with $c_p, c \in (0, 1)$. Finally, recall that $\text{tr}(\hat{\Sigma}_{\text{TME}}^2)$ and $\text{tr}(\hat{\Sigma}_{\text{SSCM}}^2)$ are understood to belong to sequences indexed by n and p , with indices suppressed for readability.

We begin with the construction used to prove Lemma S1.2 in [13]. Whenever all eigenvalues of both $\hat{\Sigma}_{\text{TME}}$ and $\hat{\Sigma}_{\text{SSCM}}$ fall inside a fixed contour $\mathcal{C}$, Cauchy's integral formula gives

$$\Pi_n = \text{tr}(\hat{\Sigma}_{\text{TME}}^2) - \text{tr}(\hat{\Sigma}_{\text{SSCM}}^2) = -\frac{1}{2\pi i} \oint_{\mathcal{C}} z^2 M_n(z) dz, \quad (\text{S1})$$

where $M_n(z) = \text{tr}[(\hat{\Sigma}_{\text{TME}} - z\mathbf{I}_p)^{-1}] - \text{tr}[(\hat{\Sigma}_{\text{SSCM}} - z\mathbf{I}_p)^{-1}]$ is the difference in the scaled Stieltjes transforms of TME and the SSCM. Here $z \in \mathbb{C}$, and we choose $\mathcal{C}$ to be the counterclockwise boundary of a rectangle with vertical sides at $x_l < x_r$ and horizontal sides at $\text{Im } z = \pm v_0$, where $v_0 > 0$.

The representation in (S1) cannot be used without modification, unfortunately. As n, p grow, realized eigenvalues of both $\hat{\Sigma}_{\text{TME}}$ and $\hat{\Sigma}_{\text{SSCM}}$ may fall outside any chosen fixed contour or approach its real-axis crossings. To control these situations, we use the regularized transform $\widehat{M}_n$ of [13]. The authors' results additionally require bounded standardized coordinates, described below. We introduce one remainder, $r_n^{(A)}$, for the contour approximation and another, $r_n^{(B)}$, for the coordinate truncation, and show that both modifications to Π_n in (S1) vanish in L^2 .

Throughout, we assume the bounded-coordinate samples are used in every object except Π_n . Likewise, let K denote a positive constant, independent of n and p (and of $z \in \mathcal{C}$), whose value may change from line to line. Define Π_n^B as the version of Π_n computed from the bounded-coordinate samples as described in the section below, with the corresponding difference $r_n^{(B)} = \Pi_n - \Pi_n^B$.

Like [15] before them, [13] adopt a regularization of $M_n(z)$.

Let $\delta_n = \varepsilon_n/n$, where $\varepsilon_n \downarrow 0$ and $\varepsilon_n > n^{-a}$ for some $a \in (0, 1)$, and define

$$\mathcal{C}_n = \{z \in \mathcal{C} : |\text{Im } z| \geq \delta_n\}.$$

For $z = x + iv \in \mathcal{C}$, we define the regularized transform $\widehat{M}_n$ as

$$\widehat{M}_n(x + iv) = \begin{cases} M_n(x + iv), & x + iv \in \mathcal{C}_n, \\ M_n(x + i\delta_n), & x \in \{x_l, x_r\}, \quad 0 \leq v < \delta_n, \\ M_n(x - i\delta_n), & x \in \{x_l, x_r\}, \quad -\delta_n < v < 0. \end{cases}$$

Thus $\widehat{M}_n$ agrees with M_n on $\mathcal{C}_n$ and holds its value constant when z approaches the real axis.

Next, define

$$\widehat{\Pi}_n^B := -\frac{1}{2\pi i} \oint_{\mathcal{C}} z^2 \widehat{M}_n(z) dz, \quad \text{so that } r_n^{(A)} = \Pi_n^B - \widehat{\Pi}_n^B.$$

Then we have

$$\Pi_n = \widehat{\Pi}_n^B + r_n^{(A)} + r_n^{(B)}.$$

To establish uniform integrability, we can show that $\sup_n \mathbb{E}|\Pi_n|^2 < \infty$. Observe that by Cauchy-Schwarz,

$$|\Pi_n|^2 = |\widehat{\Pi}_n^B + r_n^{(A)} + r_n^{(B)}|^2 \leq 3(|\widehat{\Pi}_n^B|^2 + |r_n^{(A)}|^2 + |r_n^{(B)}|^2),$$

thus, taking expectations in this inequality shows that a uniform L^2 bound for $\widehat{\Pi}_n^B$ transfers to Π_n if both remainders vanish in L^2 . We first show that $\mathbb{E}|r_n^{(A)}|^2 \rightarrow 0$ and $\mathbb{E}|r_n^{(B)}|^2 \rightarrow 0$, then establish the required uniform bound for $\widehat{\Pi}_n^B$.

A. Control of Bounded Coordinates: $r_n^{(B)}$

To control higher moments in their proofs, [13] require the coordinates of the latent random vectors allowed by their model to be bounded. Under (1), the corresponding coordinates are standard Gaussian and have moments of all orders. To apply their intermediate bounds as stated, we show that truncating these coordinates has a negligible effect on Π_n in L^2 .

Write $\mathbf{G} = \mathbf{Z}\Sigma^{1/2}$. We retain each standard Gaussian coordinate whose magnitude is at most $n^{1/3}$ and independently redraw any remaining coordinate from the standard Gaussian law conditioned on this interval. The resulting bounded coordinates satisfy Assumption 5 of [13]. Call the event $\mathcal{A}_n = \{\text{at least one entry of } \mathbf{Z} \text{ is replaced}\}$.

Standard Gaussian tail bounds plus a union bound over the np entries of the matrix $\mathbf{Z}$ give

$$P(\mathcal{A}_n) \leq 2npe^{-n^{2/3}/2}.$$

Since both Π_n and Π_n^B are based on positive semidefinite estimators with trace p , we have $|\Pi_n|, |\Pi_n^B| \leq p^2$, and hence

$$\mathbb{E}|r_n^{(B)}|^2 = \mathbb{E}|\Pi_n - \Pi_n^B|^2 \leq 4p^4 P(\mathcal{A}_n) \leq 8np^5 e^{-n^{2/3}/2} \rightarrow 0.$$

B. Control of the Regularization of $M_n(z)$: $r_n^{(A)}$

Next, observe that $\widehat{M}_n$ differs from M_n only on the four segments $\{x \pm iv : x \in \{x_l, x_r\}, 0 \leq v \leq n^{-1}\varepsilon_n\}$. Thus the effect of regularization only modifies a segment of total length $4\varepsilon_n/n$ where $\mathcal{C}$ crosses the real axis.

We begin by considering the event $\mathcal{E}_n = \{\text{all eigenvalues of } \widehat{\Sigma}_{\text{SSCM}} \text{ and } \widehat{\Sigma}_{\text{TME}} \text{ lie in } [\eta_l, \eta_r]\}$, where $x_l < \eta_l < \lambda_-(1 - \sqrt{c})^2$ and $\lambda_+(1 + \sqrt{c})^2 < \eta_r < x_r$. By Lemmas S8.4 and S1.1, $P(\mathcal{E}_n) = 1 - o(n^{-s})$ for every fixed $s > 0$. Call $d := \min(\eta_l - x_l, x_r - \eta_r) > 0$. Then because $\widehat{M}_n, M_n$ each have $2p$ terms, each with magnitude at most $1/d$ where they differ, $|M_n|, |\widehat{M}_n| \leq 2p/d$, and thus their difference is at most $4p/d$.

Then on $\mathcal{E}_n$,

$$|r_n^{(A)}| = |\Pi_n^B - \widehat{\Pi}_n^B| = \left| \frac{1}{2\pi i} \oint_{\mathcal{C}} z^2 [M_n(z) - \widehat{M}_n(z)] dz \right| \leq \frac{1}{2\pi} \max_{\mathcal{C}} |z|^2 \frac{4p}{d} \frac{4\varepsilon_n}{n} = K \frac{p\varepsilon_n}{n},$$

which gives

$$\mathbb{E}(|r_n^{(A)}|^2 \mathbf{1}\{\mathcal{E}_n\}) \leq K^2 \left(\frac{p\varepsilon_n}{n} \right)^2 \rightarrow 0.$$

Next, we evaluate the same for the complement event, $\mathcal{E}_n^C$. By construction, every evaluation of M_n used to define $\widehat{M}_n$ is at least ε_n/n from the real axis where the eigenvalues occur, so $|\widehat{M}_n| \leq 2pn/\varepsilon_n$. Then

$$\left| -\frac{1}{2\pi i} \oint_{\mathcal{C}} z^2 \widehat{M}_n(z) dz \right| = |\widehat{\Pi}_n^B| \leq \frac{1}{2\pi} \max_{\mathcal{C}} |z|^2 \frac{2pn}{\varepsilon_n} \ell(\mathcal{C}) = K \frac{pn}{\varepsilon_n},$$

with $\ell(\mathcal{C})$ denoting the (finite) length of the contour. Using the facts that $\varepsilon_n > n^{-a}$ and $p = \mathcal{O}(n)$, and reusing the fact that $|\Pi_n^B| \leq p^2$,

$$|r_n^{(A)}| = |\Pi_n^B - \widehat{\Pi}_n^B| \leq |\Pi_n^B| + |\widehat{\Pi}_n^B| \leq p^2 + K_1 \frac{pn}{\varepsilon_n} \leq K_2 n^{2+a}.$$

Then

$$\mathbb{E}(|r_n^{(A)}|^2 \mathbf{1}\{\mathcal{E}_n^C\}) \leq n^{4+2a} o(n^{-s}) \rightarrow 0,$$

for $s > 6$. Finally, we have

$$\mathbb{E}(|r_n^{(A)}|^2) = \mathbb{E}(|r_n^{(A)}|^2 \mathbf{1}\{\mathcal{E}_n\}) + \mathbb{E}(|r_n^{(A)}|^2 \mathbf{1}\{\mathcal{E}_n^C\}) \rightarrow 0.$$

Thus, $\mathbb{E}|r_n^{(A)}|^2 \rightarrow 0$ and $\mathbb{E}|r_n^{(B)}|^2 \rightarrow 0$ so it remains to show a uniform L^2 bound for $\widehat{\Pi}_n^B$.

C. Uniform L^2 Bound for $\widehat{\Pi}_n^B$

Observe that since $\oint_{\mathcal{C}} z^2 \widehat{M}_n(z_0) dz = 0$ for any fixed $z_0 \in \mathcal{C}_n$, we can represent $\widehat{\Pi}_n^B$ as a shifted integral

$$\begin{aligned} \widehat{\Pi}_n^B &= -\frac{1}{2\pi i} \oint_{\mathcal{C}} z^2 \widehat{M}_n(z) dz \\ &= -\frac{1}{2\pi i} \oint_{\mathcal{C}} z^2 \widehat{M}_n(z) dz + \frac{1}{2\pi i} \oint_{\mathcal{C}} z^2 \widehat{M}_n(z_0) dz \\ &= -\frac{1}{2\pi i} \oint_{\mathcal{C}} z^2 [\widehat{M}_n(z) - \widehat{M}_n(z_0)] dz \end{aligned}$$

Next, ([13], S1.39) and the boundedness of the fixed contour $\mathcal{C}$ imply that there exists a constant K such that

$$\sup_{n; z, z_0 \in \mathcal{C}_n} \mathbb{E} |\widehat{M}_n(z) - \widehat{M}_n(z_0)|^2 \leq K$$

For each $z \in \mathcal{C} \setminus \mathcal{C}_n$, the regularization $\widehat{M}_n(z)$ equals its value at a point in $\mathcal{C}_n$. Since $\mathcal{C}$ is fixed and bounded, the squared distances between all such evaluation points and $z_0 \in \mathcal{C}_n$ are uniformly bounded. Enlarging K if necessary, we therefore obtain

$$\sup_{\substack{z \in \mathcal{C} \\ z_0 \in \mathcal{C}_n}} \mathbb{E} |\widehat{M}_n(z) - \widehat{M}_n(z_0)|^2 \leq K,$$

uniformly for all sufficiently large n, p .

Finally, we arrive at

$$\begin{aligned} \mathbb{E} |\widehat{\Pi}_n^B|^2 &= \frac{1}{4\pi^2} \mathbb{E} \left| \oint_{\mathcal{C}} z^2 [\widehat{M}_n(z) - \widehat{M}_n(z_0)] dz \right|^2 \\ &\leq \frac{1}{4\pi^2} \mathbb{E} \left[\left(\oint_{\mathcal{C}} |z|^2 |\widehat{M}_n(z) - \widehat{M}_n(z_0)| |dz| \right)^2 \right] \\ &\leq \frac{\ell(\mathcal{C})}{4\pi^2} \mathbb{E} \left[\oint_{\mathcal{C}} |z|^4 |\widehat{M}_n(z) - \widehat{M}_n(z_0)|^2 |dz| \right] \\ &= \frac{\ell(\mathcal{C})}{4\pi^2} \oint_{\mathcal{C}} |z|^4 \mathbb{E} |\widehat{M}_n(z) - \widehat{M}_n(z_0)|^2 |dz| \\ &\leq \frac{K\ell(\mathcal{C})^2}{4\pi^2} \max_{z \in \mathcal{C}} |z|^4. \end{aligned}$$

Above, the first inequality is the triangle inequality for contour integrals. The second is Cauchy–Schwarz with respect to arc length, applied to $|z|^2 |\widehat{M}_n(z) - \widehat{M}_n(z_0)|$ and the constant function 1. Tonelli’s theorem permits the interchange of expectation and integration.

The final bound is constant and independent of n, p . Together with the decomposition $\Pi_n = \widehat{\Pi}_n^B + r_n^{(A)} + r_n^{(B)}$ and the vanishing second moments of both remainders, this yields $\sup_n \mathbb{E} |\Pi_n|^2 < \infty$. Thus $\{\Pi_n\}$ is uniformly integrable.

S2. PROOF OF PROPOSITION 2

Proposition 2 is made up of three statements, which comprise Lemma 3 and Lemma 6 below.

Lemma 3 (Frobenius Bias for SSCM). *Suppose Assumption 1 holds. Then $\|\mathbf{\Gamma}_{\text{SSCM}} - \mathbf{\Sigma}\|_F^2 = \mathcal{O}(1/p)$.*

Proof. By Assumption 1, the eigenvalues λ_k of $\mathbf{\Sigma}$ satisfy $\lambda_- \leq \lambda_k \leq \lambda_+$ for constants $\lambda_-, \lambda_+ > 0$. By rotational equivariance, $\mathbf{\Gamma}_{\text{SSCM}}$ is a function of $\mathbf{\Sigma}$ and therefore shares its eigenvectors; let ξ_k denote its eigenvalue on the eigenvector of λ_k , so that $\|\mathbf{\Gamma}_{\text{SSCM}} - \mathbf{\Sigma}\|_F^2 = \sum_k (\xi_k - \lambda_k)^2$.

$$\begin{aligned} \|\mathbf{\Gamma}_{\text{SSCM}} - \mathbf{\Sigma}\|_F^2 &= \sum_{i=1}^p (\xi_i - \lambda_i)^2 \\ &= \sum_{i=1}^p \left(\lambda_i - \frac{2}{p} \lambda_i^2 + \frac{2}{p^2} \lambda_i \left[\sum_{j=1}^p \lambda_j^2 \right] + o(1/p) - \lambda_i \right)^2 \\ &= \sum_{i=1}^p \left| -\frac{2}{p} \lambda_i^2 + \frac{2}{p^2} \lambda_i \left[\sum_{j=1}^p \lambda_j^2 \right] + o(1/p) \right|^2 \\ &\leq \sum_{i=1}^p \left(\frac{2}{p} \lambda_+^2 + \frac{2\lambda_+^3}{p} + o(1/p) \right)^2 \\ &= \sum_{i=1}^p \mathcal{O}(1/p^2) \\ &= \mathcal{O}(1/p). \end{aligned}$$

Above, the expansion of the population SSCM eigenvalues ξ_i comes from [6], who show $\|\mathbf{\Gamma}_{\text{SSCM}} - \mathbf{T}\|_{\text{op}} = o(1/p)$. The eigenvalues of $\mathbf{T}$ are provided in Equation S2.1 in the supplement of [13]. $\square$

Lemma 4 (Haar Integration of Trace Products). *Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{p \times p}$ be deterministic symmetric matrices with trace zero. Let $\mathbf{\Omega} = \mathbf{U}\mathbf{D}\mathbf{U}^\top \in \mathbb{R}^{p \times p}$ be a random matrix with $\mathbf{U}$ distributed as Haar on the orthogonal group in dimension p , with $\mathbf{D}$ diagonal, independent of $\mathbf{U}$, and $\text{tr}(\mathbf{\Omega}) = 0$ almost surely. Then $\mathbb{E}[\text{tr}(\mathbf{A}\mathbf{\Omega})\text{tr}(\mathbf{B}\mathbf{\Omega})] = \frac{2\text{tr}(\mathbf{A}\mathbf{B})\mathbb{E}\|\mathbf{\Omega}\|_F^2}{(p-1)(p+2)}$.*

Proof. This follows from expanding this product of traces and applying Lemma 2.22 in [19] to exploit symmetries of the Haar measure and convenient cancellation from the traceless matrices after simple albeit tedious manipulations. Below, δ_{ij} denotes the Kronecker delta.

First, write

$$\begin{aligned}\mathbb{E}[\text{tr}(\mathbf{A}\mathbf{\Omega})\text{tr}(\mathbf{B}\mathbf{\Omega})] &= \mathbb{E}[\text{tr}(\mathbf{A}\mathbf{U}\mathbf{D}\mathbf{U}^\top)\text{tr}(\mathbf{B}\mathbf{U}\mathbf{D}\mathbf{U}^\top)] \\ &= \mathbb{E}\left[\sum_{r,i} A_{ri}(\mathbf{U}\mathbf{D}\mathbf{U}^\top)_{ir} \sum_{\lambda,\alpha} B_{\lambda\alpha}(\mathbf{U}\mathbf{D}\mathbf{U}^\top)_{\alpha\lambda}\right] \\ &= \mathbb{E}\left[\sum_{r,i} \sum_j A_{ri} U_{ij} D_{jj} U_{rj} \sum_{\lambda,\alpha} \sum_\mu B_{\lambda\alpha} U_{\alpha\mu} D_{\mu\mu} U_{\lambda\mu}\right] \\ &= \mathbb{E}\left[\sum_{r,i,\lambda,\alpha} \sum_{j,\mu} A_{ri} U_{ij} D_{jj} U_{rj} B_{\lambda\alpha} U_{\alpha\mu} D_{\mu\mu} U_{\lambda\mu}\right].\end{aligned}$$

Then, since $\mathbf{D}$ is independent of $\mathbf{U}$,

$$\mathbb{E}[\text{tr}(\mathbf{A}\mathbf{\Omega})\text{tr}(\mathbf{B}\mathbf{\Omega})] = \sum_{r,i,\lambda,\alpha} \sum_{j,\mu} A_{ri} B_{\lambda\alpha} \mathbb{E}[D_{jj} D_{\mu\mu}] \mathbb{E}[U_{ij} U_{rj} U_{\alpha\mu} U_{\lambda\mu}]. \quad (\text{S2})$$

Applying Meckes' Lemma 2.22 with indices $j = s$ and $\beta = \mu$ in the original notation,

$$\begin{aligned}\mathbb{E}[U_{ij} U_{rj} U_{\alpha\mu} U_{\lambda\mu}] &= \frac{p+1}{(p-1)p(p+2)} [\delta_{ir}\delta_{\alpha\lambda}\delta_{jj}\delta_{\mu\mu} + \delta_{i\alpha}\delta_{r\lambda}\delta_{j\mu}\delta_{j\mu} + \delta_{i\lambda}\delta_{r\alpha}\delta_{j\mu}\delta_{j\mu}] \\ &\quad - \frac{1}{(p-1)p(p+2)} [\delta_{ir}\delta_{\alpha\lambda}\delta_{j\mu}\delta_{j\mu} + \delta_{ir}\delta_{\alpha\lambda}\delta_{j\mu}\delta_{j\mu} + \delta_{i\alpha}\delta_{r\lambda}\delta_{jj}\delta_{\mu\mu} \\ &\quad + \delta_{i\alpha}\delta_{r\lambda}\delta_{j\mu}\delta_{j\mu} + \delta_{i\lambda}\delta_{r\alpha}\delta_{jj}\delta_{\mu\mu} + \delta_{i\lambda}\delta_{r\alpha}\delta_{j\mu}\delta_{j\mu}] \\ &= \frac{p+1}{(p-1)p(p+2)} [\delta_{ir}\delta_{\alpha\lambda} + \delta_{i\alpha}\delta_{r\lambda}\delta_{j\mu} + \delta_{i\lambda}\delta_{r\alpha}\delta_{j\mu}] \\ &\quad - \frac{1}{(p-1)p(p+2)} [\delta_{ir}\delta_{\alpha\lambda}\delta_{j\mu} + \delta_{ir}\delta_{\alpha\lambda}\delta_{j\mu} + \delta_{i\alpha}\delta_{r\lambda} + \delta_{i\alpha}\delta_{r\lambda}\delta_{j\mu} + \delta_{i\lambda}\delta_{r\alpha} + \delta_{i\lambda}\delta_{r\alpha}\delta_{j\mu}] \\ &= \frac{1}{(p-1)p(p+2)} [(p+1)\delta_{ir}\delta_{\alpha\lambda} \\ &\quad + p(\delta_{i\alpha}\delta_{r\lambda}\delta_{j\mu} + \delta_{i\lambda}\delta_{r\alpha}\delta_{j\mu}) - 2(\delta_{ir}\delta_{\alpha\lambda}\delta_{j\mu}) - \delta_{i\alpha}\delta_{r\lambda} - \delta_{i\lambda}\delta_{r\alpha}].\end{aligned}$$

Plugging this back into (S2), we can take advantage of the fact that any terms that include factors $\text{tr}(\mathbf{A})$, $\text{tr}(\mathbf{B})$, or $\mathbb{E}(\text{tr}(\mathbf{D})^2)$ are zero. We start by constraining the summation to be over indices allowed to remain nonzero by the Kronecker deltas:

$$\begin{aligned}
\mathbb{E}[\text{tr}(\mathbf{A}\mathbf{\Omega})\text{tr}(\mathbf{B}\mathbf{\Omega})] &= \frac{1}{(p-1)p(p+2)} \left((p+1) \sum_{r,i,\lambda,\alpha} \sum_{j,\mu} A_{ri} B_{\lambda\alpha} \mathbb{E}[D_{jj} D_{\mu\mu}] \delta_{ir} \delta_{\alpha\lambda} \right. \\
&\quad + p \sum_{r,i,\lambda,\alpha} \sum_{j,\mu} A_{ri} B_{\lambda\alpha} \mathbb{E}[D_{jj} D_{\mu\mu}] (\delta_{i\alpha} \delta_{r\lambda} \delta_{j\mu} + \delta_{i\lambda} \delta_{r\alpha} \delta_{j\mu}) \\
&\quad - 2 \sum_{r,i,\lambda,\alpha} \sum_{j,\mu} A_{ri} B_{\lambda\alpha} \mathbb{E}[D_{jj} D_{\mu\mu}] (\delta_{ir} \delta_{\alpha\lambda} \delta_{j\mu}) - \sum_{r,i,\lambda,\alpha} \sum_{j,\mu} A_{ri} B_{\lambda\alpha} \mathbb{E}[D_{jj} D_{\mu\mu}] \delta_{i\alpha} \delta_{r\lambda} \\
&\quad \left. - \sum_{r,i,\lambda,\alpha} \sum_{j,\mu} A_{ri} B_{\lambda\alpha} \mathbb{E}[D_{jj} D_{\mu\mu}] \delta_{i\lambda} \delta_{r\alpha} \right) \\
&= \frac{1}{(p-1)p(p+2)} \left((p+1) \sum_{r=i,\lambda=\alpha} A_{ri} B_{\lambda\alpha} \sum_{j,\mu} \mathbb{E}[D_{jj} D_{\mu\mu}] \right. \\
&\quad + p \sum_{r,i,\lambda,\alpha} A_{ri} B_{\lambda\alpha} (\delta_{i\alpha} \delta_{r\lambda} + \delta_{i\lambda} \delta_{r\alpha}) \sum_{j=\mu} \mathbb{E}[D_{jj} D_{\mu\mu}] - 2 \sum_{r=i,\lambda=\alpha} A_{ri} B_{\lambda\alpha} \sum_{j=\mu} \mathbb{E}[D_{jj} D_{\mu\mu}] \\
&\quad \left. - \sum_{i=\alpha, r=\lambda,} A_{ri} B_{\lambda\alpha} \sum_{j,\mu} \mathbb{E}[D_{jj} D_{\mu\mu}] - \sum_{i=\lambda, r=\alpha} A_{ri} B_{\lambda\alpha} \sum_{j,\mu} \mathbb{E}[D_{jj} D_{\mu\mu}] \right) \\
&= \frac{1}{(p-1)p(p+2)} \left((p+1) \sum_{r=i,\lambda=\alpha} A_{ri} B_{\lambda\alpha} \mathbb{E}(\text{tr}(\mathbf{D})^2) \right. \\
&\quad + p \sum_{r,i,\lambda,\alpha} A_{ri} B_{\lambda\alpha} (\delta_{i\alpha} \delta_{r\lambda} + \delta_{i\lambda} \delta_{r\alpha}) \sum_{j=\mu} \mathbb{E}[D_{jj} D_{\mu\mu}] - 2 \text{tr}(\mathbf{A}) \text{tr}(\mathbf{B}) \sum_{j=\mu} \mathbb{E}[D_{jj} D_{\mu\mu}] \\
&\quad \left. - \sum_{i=\alpha, r=\lambda,} A_{ri} B_{\lambda\alpha} \mathbb{E}(\text{tr}(\mathbf{D})^2) - \sum_{i=\lambda, r=\alpha} A_{ri} B_{\lambda\alpha} \mathbb{E}(\text{tr}(\mathbf{D})^2) \right) \\
&= \frac{1}{(p-1)p(p+2)} \left(p \sum_{r,i,\lambda,\alpha} A_{ri} B_{\lambda\alpha} (\delta_{i\alpha} \delta_{r\lambda} + \delta_{i\lambda} \delta_{r\alpha}) \sum_j \mathbb{E}[D_{jj}^2] \right) \\
&= \frac{2}{(p-1)(p+2)} \left(\sum_{r,i} A_{ri} B_{ri} \sum_j \mathbb{E}[D_{jj}^2] \right) \\
&= \frac{2 \text{tr}(\mathbf{A}\mathbf{B}) \mathbb{E} \|\mathbf{\Omega}\|_F^2}{(p-1)(p+2)}.
\end{aligned}$$

□

Lemma 5 (Second-Moment Control for TME). *Let $\hat{\Sigma}_{TME}$ and $\hat{\Sigma}_{SSCM}$ denote Tyler's M-estimator and the SSCM, respectively, for data generated according to (1), with Assumption 1 of this paper satisfied and $c_p, c \in (0, 1)$. Then $\text{tr}(\mathbb{E}(\hat{\Sigma}_{TME}^2)) = \mathcal{O}(p)$.*

Proof. Observe that $\text{tr}(\mathbb{E}(\hat{\Sigma}_{TME}^2)) = \mathbb{E}\Pi_n + \text{tr}(\mathbb{E}(\hat{\Sigma}_{SSCM}^2))$. By Proposition 1, we have $\mathbb{E}\Pi_n = \mathcal{O}(1)$. By the exact identity given in the proof of Lemma 2(a) and the characterization of the eigenvalues of $\mathbf{\Gamma}_{SSCM}$ provided by Lemma 3, we have

$$\begin{aligned}
\mathbb{E} \text{tr}(\hat{\Sigma}_{SSCM}^2) &= pc_p + \left(1 - \frac{1}{n}\right) \text{tr}(\mathbf{\Gamma}_{SSCM}^2) \\
&= pc_p + \left(1 - \frac{1}{n}\right) \left(\text{tr}(\mathbf{\Sigma}^2) + \mathcal{O}(1) \right) \\
&\leq pc_p + \left(1 - \frac{1}{n}\right) \left(p\lambda_+^2 + \mathcal{O}(1) \right) \\
&= \mathcal{O}(p).
\end{aligned}$$

Thus, $\text{tr}(\mathbb{E}(\hat{\Sigma}_{TME}^2)) = \mathcal{O}(1) + \mathcal{O}(p) = \mathcal{O}(p)$.

□

Lemma 6 (Frobenius Bias and Cross-Term Control for TME). *Assume Tyler's M-estimator exists, and Assumption 1 holds with $c_p, c < 1$. Then*

- a) $\|\mathbb{E}(\hat{\Sigma}_{TME}) - \mathbf{\Sigma}\|_F^2 = \mathcal{O}(1/p)$ and
- b) $\text{tr}(\mathbb{E}(\hat{\Sigma}_{TME})\mathbf{\Sigma}) = \text{tr}(\mathbf{\Sigma}^2) + \mathcal{O}(1/p)$.

Proof. Proving both statements relies on recasting quantities as traceless matrices and leveraging Lemma 4 for control of the trace products involved.

First, it is convenient to leverage the affine equivariance and distribution-free properties of Tyler's M-estimator for shape to write $\hat{\Sigma}_{\text{TME}}(\mathbf{X})$ as a function of the estimator applied to the whitened data matrix $\mathbf{Z} := \mathbf{G}\Sigma^{-1/2}$. Write

$$\hat{\Sigma}_{\text{TME}}(\mathbf{X}) = \frac{\Sigma^{1/2} \hat{\Sigma}_0 \Sigma^{1/2}}{\frac{1}{p} \text{tr}(\Sigma \hat{\Sigma}_0)},$$

where $\hat{\Sigma}_0 := \hat{\Sigma}_{\text{TME}}(\mathbf{Z})$.

As a simple first case, consider when the true shape is spherical: $\Sigma = \mathbf{I}$.

Because Tyler's M-estimator is affine equivariant for shape and the distribution of $\mathbf{Z}$ is rotationally invariant, we must have $\mathbb{E}(\hat{\Sigma}_0) = \mathbf{O} \mathbb{E}(\hat{\Sigma}_0) \mathbf{O}^\top$ for any orthogonal matrix $\mathbf{O} \in \mathbb{R}^{p \times p}$. By Schur's Lemma, the only matrix that is invariant under congruence transforms with any orthogonal matrix is a multiple of the identity, so $\mathbb{E}(\hat{\Sigma}_0) \propto \mathbf{I}$. Since $\text{tr}(\hat{\Sigma}_0) = p$ a.s., this means TME is exactly unbiased when $\Sigma = \mathbf{I}$.

Next, we extend to the case of a non-spherical shape matrix.

Define the matrices $\Delta := \Sigma - \mathbf{I}$, $\hat{\mathbf{E}} := \hat{\Sigma}_0 - \mathbf{I}$, $\mathbf{B} := \Sigma^2 - p^{-1} \text{tr}(\Sigma^2) \mathbf{I}$, and the scalar $u := p^{-1} \text{tr}(\Delta \hat{\mathbf{E}})$. By the uniform bounds on the eigenvalues of Σ , the squared Frobenius norms of the deterministic traceless matrices Δ and $\mathbf{B}$ are $\mathcal{O}(p)$. Additionally, by Lemma 5 and the fact that $\text{tr}(\hat{\Sigma}_0) = p$, we have $\mathbb{E} \|\hat{\mathbf{E}}\|_F^2 = \mathcal{O}(p)$.

By construction, $\text{tr}(\Delta) = \text{tr}(\hat{\mathbf{E}}) = \text{tr}(\mathbf{B}) = 0$ almost surely, and $\mathbb{E}(\hat{\mathbf{E}}) = \mathbf{0}$. Likewise, $\mathbb{E}(u) = 0$. Recall that from the Assumption 1, Σ has uniformly bounded eigenvalues $\lambda_- \leq \lambda_k(\Sigma) \leq \lambda_+$ for $k = 1, \dots, p$. Then we have λ_- as a lower bound on $1 + u = p^{-1} \text{tr}(\Sigma \hat{\Sigma}_0)$:

$$\begin{aligned} \lambda_- &= \frac{\lambda_-}{p} \text{tr}(\hat{\Sigma}_0) = \frac{1}{p} \text{tr}(\lambda_- \hat{\Sigma}_0) \\ &\leq \frac{1}{p} \text{tr}(\Sigma \hat{\Sigma}_0) \\ &= \frac{1}{p} \text{tr}((\Delta + \mathbf{I})(\hat{\mathbf{E}} + \mathbf{I})) \\ &= 1 + u. \end{aligned} \tag{S3}$$

Finally, because $\hat{\Sigma}_0$ has a rotationally invariant distribution, its eigenvectors are Haar-distributed on the orthogonal group in dimension p and independent of its eigenvalues [20]. Thus, because $\hat{\mathbf{E}} = \hat{\Sigma}_0 - \mathbf{I}$, it shares these same Haar-distributed eigenvectors, while its diagonal matrix of eigenvalues is simply shifted by the identity. Consequently, $\mathbb{E}(u^2) = \frac{1}{p^2} \mathbb{E}(\text{tr}(\Delta \hat{\mathbf{E}}) \text{tr}(\Delta \hat{\mathbf{E}})) = \frac{1}{p^2} \frac{2 \|\Delta\|_F^2 \mathbb{E} \|\hat{\mathbf{E}}\|_F^2}{(p-1)(p+2)} = \mathcal{O}(1/p^2)$ immediately by Lemma 4.

a) Starting with $\|\mathbb{E}(\hat{\Sigma}_{\text{TME}}) - \Sigma\|_F^2 = \mathcal{O}(1/p)$.

First we use the unbiasedness of $\hat{\Sigma}_0$ to write both terms as expectations since $\mathbb{E}[\Sigma^{1/2} \hat{\Sigma}_0 \Sigma^{1/2}] = \Sigma$:

$$\begin{aligned} \|\mathbb{E}(\hat{\Sigma}_{\text{TME}}) - \Sigma\|_F^2 &= \left\| \mathbb{E} \left[\frac{\Sigma^{1/2} \hat{\Sigma}_0 \Sigma^{1/2}}{\frac{1}{p} \text{tr}(\Sigma \hat{\Sigma}_0)} \right] - \mathbb{E}[\Sigma^{1/2} \hat{\Sigma}_0 \Sigma^{1/2}] \right\|_F^2 \\ &= \left\| \mathbb{E} \left[\frac{\Sigma^{1/2} \hat{\Sigma}_0 \Sigma^{1/2}}{\frac{1}{p} \text{tr}(\Sigma \hat{\Sigma}_0)} - \Sigma^{1/2} \hat{\Sigma}_0 \Sigma^{1/2} \right] \right\|_F^2 \\ &= \left\| \mathbb{E} \left[\Sigma^{1/2} \hat{\Sigma}_0 \Sigma^{1/2} \left(\frac{1}{\frac{1}{p} \text{tr}(\Sigma \hat{\Sigma}_0)} - 1 \right) \right] \right\|_F^2. \end{aligned}$$

Next, using (S3) to write the trace in the denominator in terms of u :

$$\begin{aligned}
\|\mathbb{E}(\hat{\Sigma}_{\text{TME}}) - \Sigma\|_F^2 &= \left\| \mathbb{E} \left[\Sigma^{1/2} \hat{\Sigma}_0 \Sigma^{1/2} \left(\frac{1}{1+u} - 1 \right) \right] \right\|_F^2 \\
&= \left(\left\| \mathbb{E} \left[\Sigma^{1/2} \hat{\Sigma}_0 \Sigma^{1/2} \left(\frac{-u}{1+u} \right) \right] \right\|_F \right)^2 \\
&\leq \left(\mathbb{E} \left\| \Sigma^{1/2} \hat{\Sigma}_0 \Sigma^{1/2} \left(\frac{-u}{1+u} \right) \right\|_F \right)^2 \quad (*) \\
&= \left(\mathbb{E} \left\{ \left\| \Sigma^{1/2} \hat{\Sigma}_0 \Sigma^{1/2} \right\|_F \left| \frac{u}{1+u} \right| \right\} \right)^2 \\
&\leq \mathbb{E} \|\Sigma^{1/2} \hat{\Sigma}_0 \Sigma^{1/2}\|_F^2 \mathbb{E} \left[\left(\frac{u}{1+u} \right)^2 \right] \quad (**) \\
&\leq \mathbb{E} \|\Sigma^{1/2} \hat{\Sigma}_0 \Sigma^{1/2}\|_F^2 \frac{1}{\lambda_-^2} \mathbb{E}[u^2] \\
&\leq \frac{1}{\lambda_-^2} \|\Sigma\|_{\text{op}}^2 \mathbb{E} \|\hat{\Sigma}_0\|_F^2 \mathbb{E}[u^2] \\
&\leq \frac{\lambda_+^2}{\lambda_-^2} \text{tr}(\mathbb{E}(\hat{\Sigma}_0^2)) \mathbb{E}[u^2] \\
&= \mathcal{O}(p) \mathcal{O}(1/p^2) \\
&= \mathcal{O}(1/p).
\end{aligned}$$

Where $(*)$ follows from Jensen's Inequality and $(**)$ from the Cauchy-Schwarz inequality. The inequality that follows on the subsequent line is a direct consequence of (S3) and its bound $1+u \geq \lambda_-$, and $\text{tr}(\mathbb{E}(\hat{\Sigma}_0^2)) = \mathcal{O}(p)$ is shown in Lemma 5.

b) Next, $\text{tr}(\mathbb{E}(\hat{\Sigma}_{\text{TME}})\Sigma) = \text{tr}(\Sigma^2) + \mathcal{O}(1/p)$

$$\begin{aligned}
\text{tr}(\hat{\Sigma}_{\text{TME}}\Sigma) &= \frac{\text{tr}(\Sigma^{1/2} \hat{\Sigma}_0 \Sigma^{1/2} \Sigma)}{\frac{1}{p} \text{tr}(\Sigma \hat{\Sigma}_0)} \\
&= \frac{\text{tr}(\Sigma^2 \hat{\Sigma}_0)}{\frac{1}{p} \text{tr}(\Sigma \hat{\Sigma}_0)} \\
&= \frac{\text{tr}((\mathbf{B} + \frac{1}{p} \text{tr}(\Sigma^2) \mathbf{I})(\hat{\mathbf{E}} + \mathbf{I}))}{\frac{1}{p} \text{tr}(\Sigma \hat{\Sigma}_0)} \\
&= \frac{1}{\frac{1}{p} \text{tr}(\Sigma \hat{\Sigma}_0)} \text{tr}[(\mathbf{B} \hat{\mathbf{E}}) + \mathbf{B} + \frac{1}{p} \text{tr}(\Sigma^2) \hat{\mathbf{E}} + \frac{1}{p} \text{tr}(\Sigma^2) \mathbf{I}] \\
&= \frac{\text{tr}(\mathbf{B} \hat{\mathbf{E}})}{\frac{1}{p} \text{tr}(\Sigma \hat{\Sigma}_0)} + \frac{\text{tr}(\Sigma^2)}{\frac{1}{p} \text{tr}(\Sigma \hat{\Sigma}_0)} \\
&= \frac{1}{1+u} (\text{tr}(\mathbf{B} \hat{\mathbf{E}}) + \text{tr}(\Sigma^2)) \\
&= \left(1 - u + \frac{u^2}{1+u} \right) (\text{tr}(\mathbf{B} \hat{\mathbf{E}}) + \text{tr}(\Sigma^2)) \\
&= \text{tr}(\Sigma^2) + \text{tr}(\mathbf{B} \hat{\mathbf{E}}) - u \text{tr}(\Sigma^2) - u \text{tr}(\mathbf{B} \hat{\mathbf{E}}) + \frac{u^2}{1+u} [\text{tr}(\Sigma^2) + \text{tr}(\mathbf{B} \hat{\mathbf{E}})].
\end{aligned}$$

Taking absolute deviations of expectations, since $\mathbb{E} \text{tr}(\mathbf{B} \hat{\mathbf{E}}) = 0$ and $\mathbb{E}(u \text{tr}(\Sigma^2)) = 0$, we can write

$$\begin{aligned}
|\text{tr}(\mathbb{E}(\hat{\Sigma}_{\text{TME}})\Sigma) - \text{tr}(\Sigma^2)| &\leq 0 + 0 + |\mathbb{E}(u\text{tr}(\mathbf{B}\hat{\mathbf{E}}))| + \left| \mathbb{E}\left[\frac{u^2}{1+u}[\text{tr}(\Sigma^2) + \text{tr}(\mathbf{B}\hat{\mathbf{E}})]\right] \right| \\
&= \left| \frac{1}{p}\mathbb{E}[\text{tr}(\Delta\hat{\mathbf{E}})\text{tr}(\mathbf{B}\hat{\mathbf{E}})] \right| + \left| \mathbb{E}\left[\frac{u^2}{1+u}[\text{tr}(\Sigma^2) + \text{tr}(\mathbf{B}\hat{\mathbf{E}})]\right] \right| \\
&\leq \left| \frac{1}{p}\mathbb{E}[\text{tr}(\Delta\hat{\mathbf{E}})\text{tr}(\mathbf{B}\hat{\mathbf{E}})] \right| + \left| \frac{1}{\lambda_-}\mathbb{E}\left[u^2[\text{tr}(\Sigma^2) + \text{tr}(\mathbf{B}\hat{\mathbf{E}})]\right] \right| \\
&= \left| \frac{1}{p}\frac{2\text{tr}(\Delta\mathbf{B})\mathbb{E}\|\hat{\mathbf{E}}\|_F^2}{(p-1)(p+2)} \right| + \left| \frac{1}{\lambda_-}\mathbb{E}\left[u^2[\text{tr}(\Sigma^2) + \text{tr}((\Sigma^2 - \frac{1}{p}\text{tr}(\Sigma^2)\mathbf{I})\hat{\mathbf{E}})]\right] \right| \\
&= \left| \frac{1}{p}\frac{2\text{tr}(\Delta\mathbf{B})\mathbb{E}\|\hat{\mathbf{E}}\|_F^2}{(p-1)(p+2)} \right| + \left| \frac{1}{\lambda_-}\mathbb{E}\left[u^2[\text{tr}(\Sigma^2) + \text{tr}(\Sigma^2\hat{\mathbf{E}})]\right] \right| \\
&= \left| \frac{1}{p}\frac{2\text{tr}(\Delta\mathbf{B})\mathbb{E}\|\hat{\mathbf{E}}\|_F^2}{(p-1)(p+2)} \right| + \left| \frac{1}{\lambda_-}\mathbb{E}\left[u^2[\text{tr}(\Sigma^2 + \Sigma^2(\hat{\Sigma}_0 - \mathbf{I}))]\right] \right| \\
&= \left| \frac{1}{p}\frac{2\text{tr}(\Delta\mathbf{B})\mathbb{E}\|\hat{\mathbf{E}}\|_F^2}{(p-1)(p+2)} \right| + \left| \frac{1}{\lambda_-}\mathbb{E}[u^2\text{tr}(\Sigma^2\hat{\Sigma}_0)] \right| \\
&\leq \left| \frac{1}{p}\frac{2\text{tr}(\Delta\mathbf{B})\mathbb{E}\|\hat{\mathbf{E}}\|_F^2}{(p-1)(p+2)} \right| + \frac{1}{\lambda_-}\mathbb{E}[u^2\lambda_+^2\text{tr}(\hat{\Sigma}_0)] \\
&= \left| \frac{1}{p}\frac{2\text{tr}(\Delta\mathbf{B})\mathbb{E}\|\hat{\mathbf{E}}\|_F^2}{(p-1)(p+2)} \right| + \frac{\lambda_+^2 p}{\lambda_-}\mathbb{E}[u^2] \\
&= \frac{1}{p}\frac{2\mathcal{O}(p)\mathcal{O}(p)}{(p-1)(p+2)} + \frac{\lambda_+^2 p}{\lambda_-}\mathcal{O}(1/p^2) \\
&= \mathcal{O}(1/p),
\end{aligned}$$

where the penultimate line uses the fact that $|\text{tr}(\Delta\mathbf{B})| \leq \|\Delta\|_F\|\mathbf{B}\|_F = \mathcal{O}(\sqrt{p})\mathcal{O}(\sqrt{p}) = \mathcal{O}(p)$.

Thus we have $\text{tr}(\mathbb{E}(\hat{\Sigma}_{\text{TME}})\Sigma) = \text{tr}(\Sigma^2) + \mathcal{O}(1/p)$. □

S3. SIMULATION TABLES

For ease of comparison across varying p , in the tables below, theoretical risk is reported as $p^{-1}\mathbb{E}\|\hat{\Sigma} - \Sigma\|_F^2$, and simulated risk is reported as the average similarly scaled risk, $(Rp)^{-1} \sum_{r=1}^R \|\hat{\Sigma}^{(r)} - \Sigma\|_F^2$, over $R = 150$ simulations, with $\hat{\Sigma}^{(r)}$ the r th replicate. Monte Carlo standard errors, in parentheses, are standard errors of the mean over replicates; because the theoretical risk is a constant, the same standard error applies to Sim. – Theory. The SSCM’s risk improvement relative to a benchmark is computed from paired replicates, with a delta-method standard error. Theoretical values are the approximations of the high-dimensional limits without finite- p bias correction. Aspect ratios $c_p = p/n$ are displayed to one decimal place; theoretical risks are calculated using the unrounded ratios.

TABLE I
RISK AND RUNTIME BY SHAPE ESTIMATOR, $p = 25$

Estimator	Sim. Risk (MC SE)	Theory Risk	Sim. – Theory	Runtime (ms)	SSCM Risk Improv. % (SE)	Runtime vs SSCM
$c_p = 0.1, \rho = 8$						
SSCM	0.1062 (0.0013)	0.0906	0.0157	0.08	-	-
Latent SCM	0.1082 (0.0018)	0.1094	-0.0013	0.03	1.8 (1.1)	-58.5%
Tyler's M	0.1155 (0.0019)	0.1183	-0.0029	1.39	8.01 (0.81)	1712.6%
$c_p = 0.1, \rho = 100$						
SSCM	0.0838 (0.0025)	0.0815	0.0023	0.10	-	-
Latent SCM	0.1183 (0.0038)	0.1185	-0.0002	0.04	29.1 (1.6)	-58.1%
Tyler's M	0.1200 (0.0039)	0.1274	-0.0074	1.93	30.2 (1.2)	1928.6%
$c_p = 0.1, \rho = 10000$						
SSCM	0.0782 (0.0029)	0.0800	-0.0018	0.07	-	-
Latent SCM	0.1194 (0.0043)	0.1200	-0.0006	0.03	34.5 (1.8)	-57.4%
Tyler's M	0.1206 (0.0042)	0.1289	-0.0082	1.38	35.2 (1.2)	1825.1%
$c_p = 0.5, \rho = 8$						
SSCM	0.4753 (0.0059)	0.4528	0.0226	0.04	-	-
Latent SCM	0.5534 (0.0097)	0.5472	0.0062	0.01	14.1 (1.4)	-64.5%
Tyler's M	0.623 (0.011)	0.627	-0.005	2.12	23.65 (0.79)	4987.3%
$c_p = 0.5, \rho = 100$						
SSCM	0.435 (0.011)	0.408	0.027	0.04	-	-
Latent SCM	0.608 (0.018)	0.592	0.016	0.01	28.6 (1.5)	-63.4%
Tyler's M	0.664 (0.019)	0.672	-0.008	2.12	34.6 (1.4)	5163.7%
$c_p = 0.5, \rho = 10000$						
SSCM	0.409 (0.013)	0.400	0.009	0.04	-	-
Latent SCM	0.632 (0.022)	0.600	0.032	0.01	35.2 (2.0)	-63.3%
Tyler's M	0.676 (0.025)	0.680	-0.004	2.27	39.5 (1.7)	5492.9%
$c_p = 0.9, \rho = 8$						
SSCM	0.832 (0.010)	0.809	0.024	0.04	-	-
Latent SCM	0.962 (0.018)	0.977	-0.015	0.01	13.5 (1.4)	-61.4%
Tyler's M	1.526 (0.028)	1.644	-0.117	11.55	45.48 (0.98)	30803.4%
$c_p = 0.9, \rho = 100$						
SSCM	0.749 (0.020)	0.728	0.021	0.04	-	-
Latent SCM	1.039 (0.038)	1.058	-0.019	0.01	27.9 (2.0)	-65.8%
Tyler's M	1.659 (0.056)	1.725	-0.066	11.28	54.9 (1.5)	29201.6%
$c_p = 0.9, \rho = 10000$						
SSCM	0.678 (0.019)	0.714	-0.036	0.04	-	-
Latent SCM	1.015 (0.035)	1.071	-0.057	0.01	33.2 (2.2)	-65.1%
Tyler's M	1.601 (0.057)	1.738	-0.137	11.85	57.6 (1.5)	29727.0%

TABLE II
RISK AND RUNTIME BY SHAPE ESTIMATOR, $p = 50$

Estimator	Sim. Risk (MC SE)	Theory Risk	Sim. – Theory	Runtime (ms)	SSCM Risk Improv. % (SE)	Runtime vs SSCM
$c_p = 0.1, \rho = 8$						
SSCM	0.10114 (0.00073)	0.09528	0.00586	0.24	-	-
Latent SCM	0.10691 (0.00089)	0.10472	0.00219	0.12	5.40 (0.43)	-47.2%
Tyler's M	0.11083 (0.00093)	0.10917	0.00166	4.09	8.74 (0.28)	1628.8%
$c_p = 0.1, \rho = 100$						

TABLE III
RISK AND RUNTIME BY SHAPE ESTIMATOR, $p = 100$ (continued)

Estimator	Sim. Risk (MC SE)	Theory Risk	Sim. — Theory	Runtime (ms)	SSCM Risk Improv. % (SE)	Runtime vs SSCM
SSCM	0.4773 (0.0042)	0.4750	0.0023	0.23	-	-
Latent SCM	0.5296 (0.0051)	0.5250	0.0046	0.16	9.87 (0.47)	-33.0%
Tyler's M	0.5431 (0.0048)	0.5450	-0.0019	14.48	12.12 (0.32)	6108.7%
$c_p = 0.9, \rho = 8$						
SSCM	0.8837 (0.0032)	0.8796	0.0040	0.16	-	-
Latent SCM	0.9268 (0.0054)	0.9222	0.0047	0.10	4.66 (0.39)	-38.8%
Tyler's M	1.1065 (0.0048)	1.1040	0.0026	62.37	20.14 (0.24)	38984.3%
$c_p = 0.9, \rho = 100$						
SSCM	0.8564 (0.0065)	0.8592	-0.0028	0.16	-	-
Latent SCM	0.9394 (0.0082)	0.9426	-0.0032	0.10	8.83 (0.52)	-39.0%
Tyler's M	1.1137 (0.0085)	1.1244	-0.0107	61.13	23.11 (0.43)	38798.6%
$c_p = 0.9, \rho = 10000$						
SSCM	0.8561 (0.0069)	0.8559	0.0002	0.16	-	-
Latent SCM	0.9553 (0.0089)	0.9459	0.0094	0.10	10.38 (0.53)	-38.0%
Tyler's M	1.1264 (0.0090)	1.1277	-0.0013	62.46	24.00 (0.48)	38698.4%

TABLE IV
RISK AND RUNTIME BY SHAPE ESTIMATOR, $p = 1000$

Estimator	Sim. Risk (MC SE)	Theory Risk	Sim. — Theory	Runtime (ms)	SSCM Risk Improv. % (SE)	Runtime vs SSCM
$c_p = 0.1, \rho = 8$						
SSCM	0.099721 (0.000040)	0.099764	-0.000043	584.39	-	-
Latent SCM	0.100202 (0.000043)	0.100236	-0.000034	496.04	0.480 (0.011)	-15.1%
Tyler's M	0.100404 (0.000041)	0.100458	-0.000054	11769.15	0.6807 (0.0032)	1913.9%
$c_p = 0.1, \rho = 100$						
SSCM	0.099584 (0.000078)	0.099538	0.000047	568.09	-	-
Latent SCM	0.100536 (0.000080)	0.100462	0.000073	479.57	0.946 (0.015)	-15.6%
Tyler's M	0.100723 (0.000079)	0.100685	0.000038	11565.28	1.1302 (0.0095)	1935.8%
$c_p = 0.1, \rho = 10000$						
SSCM	0.099559 (0.000084)	0.099500	0.000058	573.55	-	-
Latent SCM	0.100590 (0.000089)	0.100500	0.000091	484.30	1.026 (0.021)	-15.6%
Tyler's M	0.100776 (0.000085)	0.100722	0.000054	11571.41	1.208 (0.010)	1917.5%
$c_p = 0.5, \rho = 8$						
SSCM	0.49881 (0.00018)	0.49882	-0.00001	110.14	-	-
Latent SCM	0.50147 (0.00022)	0.50118	0.00029	100.14	0.529 (0.026)	-9.1%
Tyler's M	0.50315 (0.00018)	0.50318	-0.00003	6576.61	0.8614 (0.0041)	5871.2%
$c_p = 0.5, \rho = 100$						
SSCM	0.49740 (0.00038)	0.49769	-0.00029	110.04	-	-
Latent SCM	0.50243 (0.00042)	0.50231	0.00012	99.98	1.002 (0.038)	-9.1%
Tyler's M	0.50401 (0.00038)	0.50431	-0.00030	6532.00	1.312 (0.011)	5836.0%
$c_p = 0.5, \rho = 10000$						
SSCM	0.49719 (0.00040)	0.49750	-0.00031	112.45	-	-
Latent SCM	0.50247 (0.00046)	0.50250	-0.00002	101.83	1.051 (0.036)	-9.4%
Tyler's M	0.50410 (0.00040)	0.50450	-0.00040	6740.00	1.370 (0.012)	5893.6%
$c_p = 0.9, \rho = 8$						
SSCM	0.89801 (0.00033)	0.89796	0.00005	63.02	-	-
Latent SCM	0.90253 (0.00045)	0.90222	0.00031	56.83	0.500 (0.032)	-9.8%
Tyler's M	0.92039 (0.00035)	0.92023	0.00015	22801.89	2.431 (0.012)	36082.4%
$c_p = 0.9, \rho = 100$						
SSCM	0.89604 (0.00073)	0.89593	0.00011	64.32	-	-
Latent SCM	0.90372 (0.00085)	0.90425	-0.00054	57.88	0.850 (0.050)	-10.0%
Tyler's M	0.92271 (0.00074)	0.92227	0.00044	22955.86	2.891 (0.019)	35590.4%
$c_p = 0.9, \rho = 10000$						
SSCM	0.89618 (0.00066)	0.89559	0.00059	62.94	-	-
Latent SCM	0.90552 (0.00082)	0.90459	0.00094	56.87	1.031 (0.046)	-9.6%
Tyler's M	0.92338 (0.00067)	0.92260	0.00078	22019.33	2.945 (0.023)	34885.3%